

HCDE 511

Information Visualization

Winter 2026

Week
3



Week

Complete A2 peer review

Short break

Fundamental Graphs

Break

Finding data

Data formats

Tool intros

Data cleaning workshop(s)

Short break

Assignment 3 overview

Group time (1hr +)

Week 1

Intro
Data
encoding &
mapping

2

Graphical
excellence
& integrity
Critiquing

3

Fundamen-
tal graphs
Working
with data

4

Visual
perception
& design

5

Tasks &
inter-
actions

6

Collabora-
tive
critique

7

Multi-
variate
analysis

8

Trees &
networks

9

Animation,
context,
story-
telling

10

Final pre-
sentations

Finals

A1 Good/bad

A2 Static vis

A3 Exploratory analysis

Project proposal

Rev. prop.

Final project deliverable

Final project writeup

Peer Review:

**Grade your classmate's
assignment 2 submission**

**If canvas has not assigned
one, let us know**

- 1** List the source of the visualization (a bibliographic citation is preferred, including the date if you can determine it). Include the creator and what you know of their background. **1pt**
- 2** How does this visualization relate to your final project topic? **2pts**
- 3** Who is the intended audience? What is its intended goal or purpose? **2pts**
- 4** What information does this visualization represent? **2pts**
- 5** How many data dimensions does it encode? Are the encoding mappings appropriate? **2pts**
- 6** List several tasks, comparisons or evaluations it enables. **2pts**
- 7** Which of Tufte's principles of excellence best describe why it is good? **2pts**
- 8** List at least three strengths and three weaknesses of the visualization. **2pts**
- 9** Does the visualization serve its intended purpose, in your opinion? **2pts**
- 10** Can you suggest any improvements? **1pts**
- 11** Why do you like this visualization? **2pts**

Break

until x:xx pm

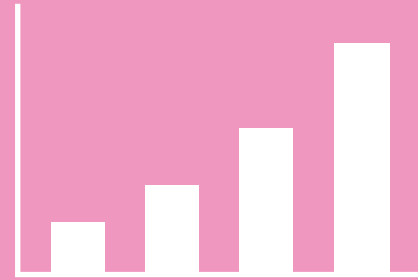
Fundamental Graphs



**Nominal
comparison**



Time series



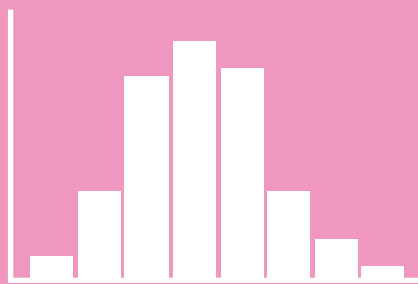
Ranking



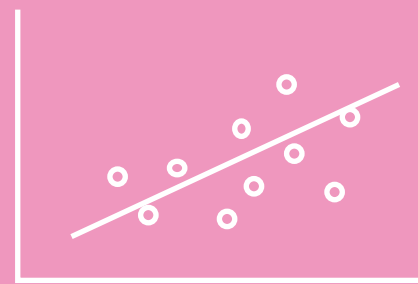
Part-to-whole



Deviation



Distribution

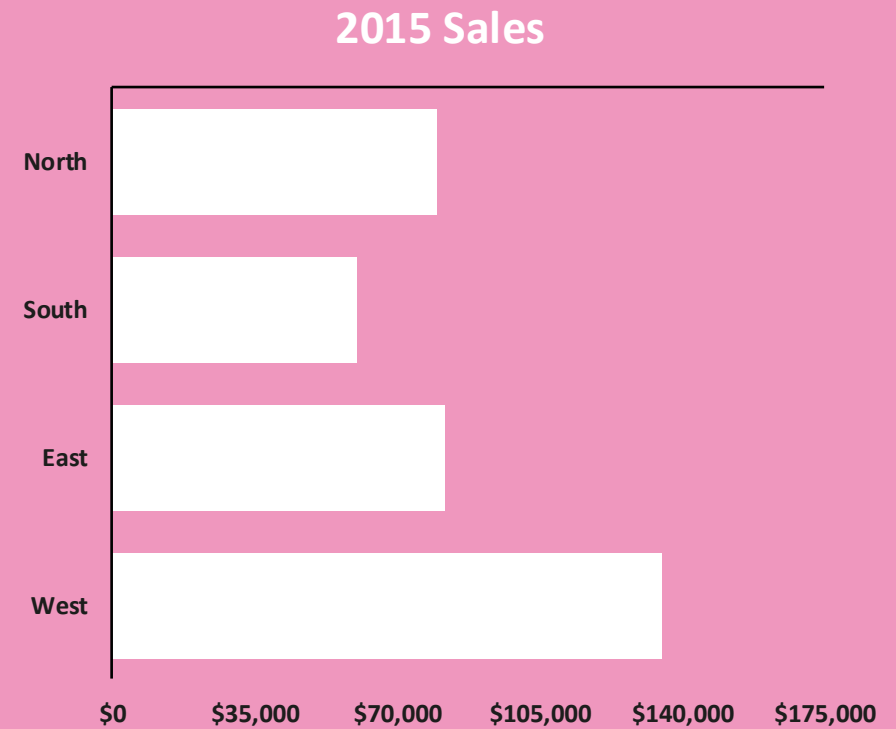
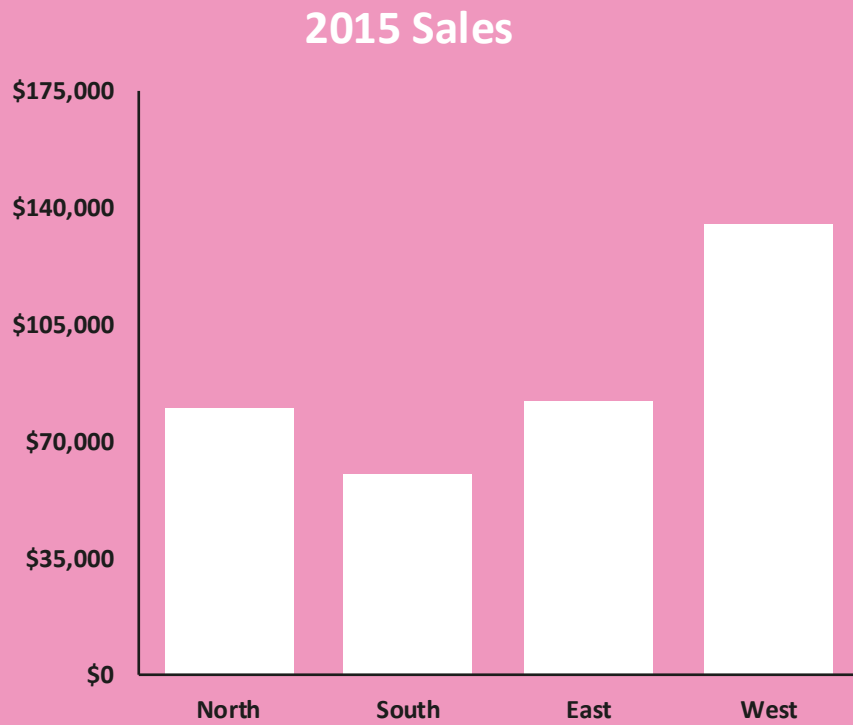


Correlation

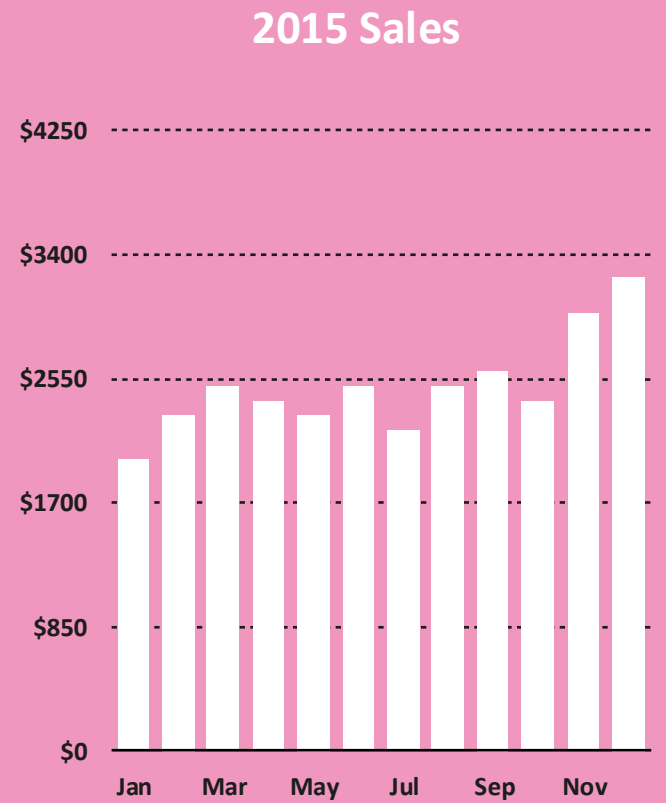
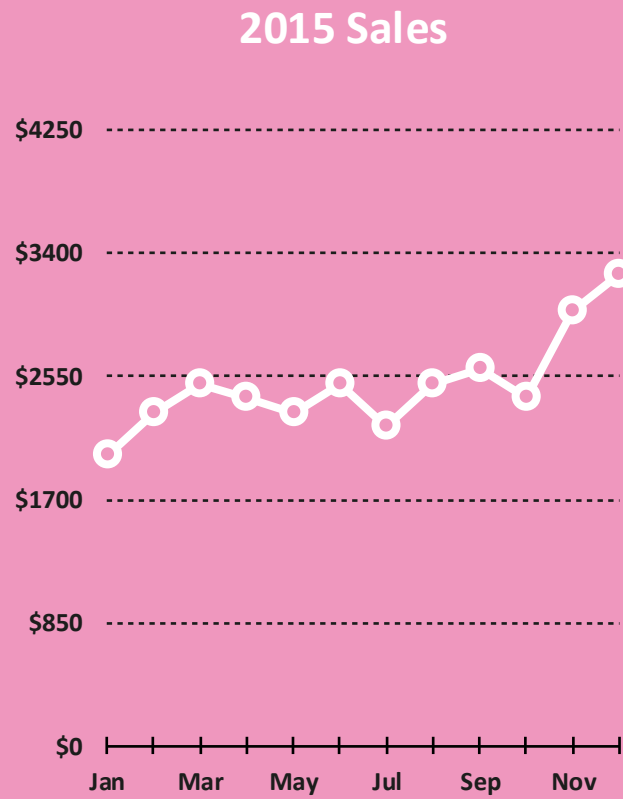
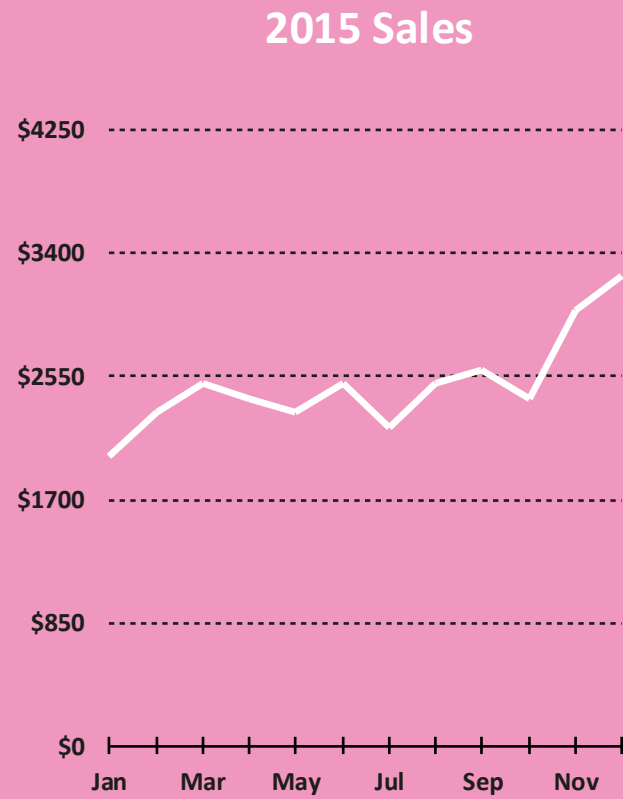


Geospatial

Nominal Comparison



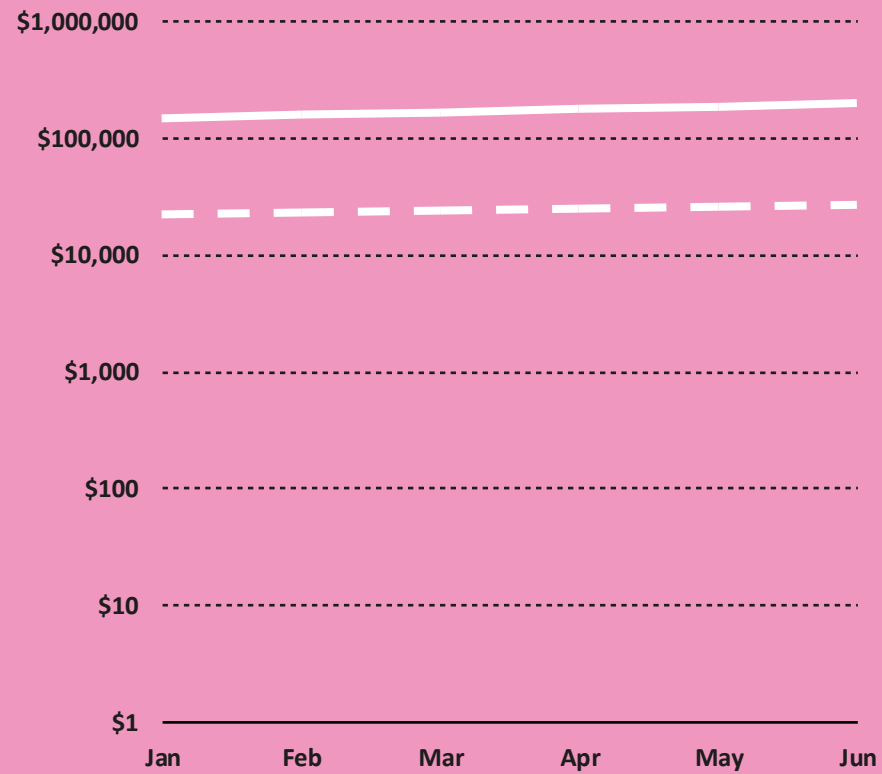
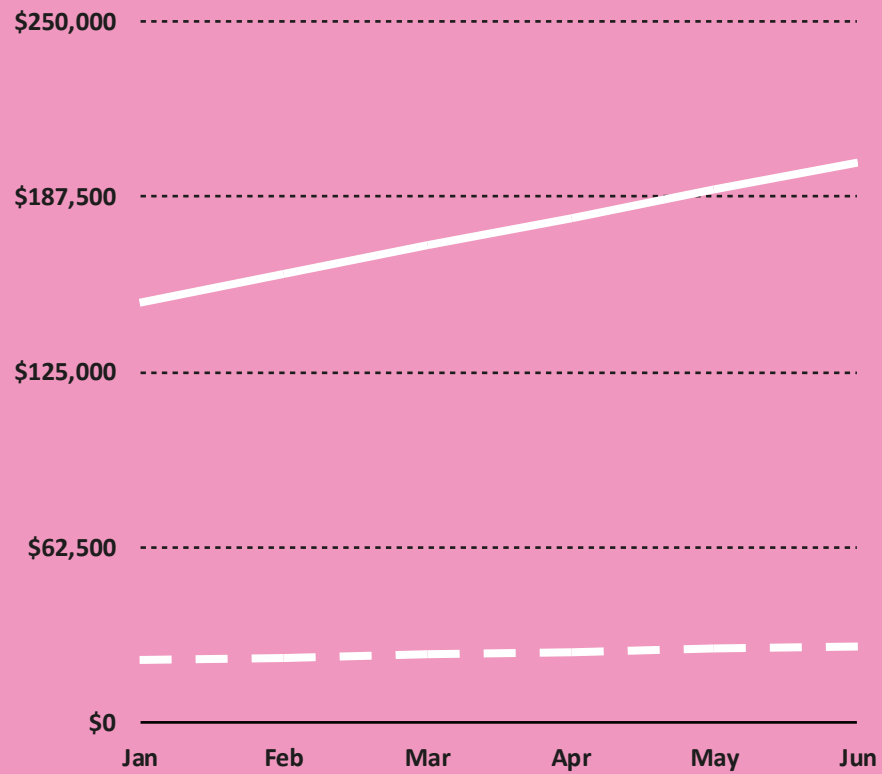
Time Series



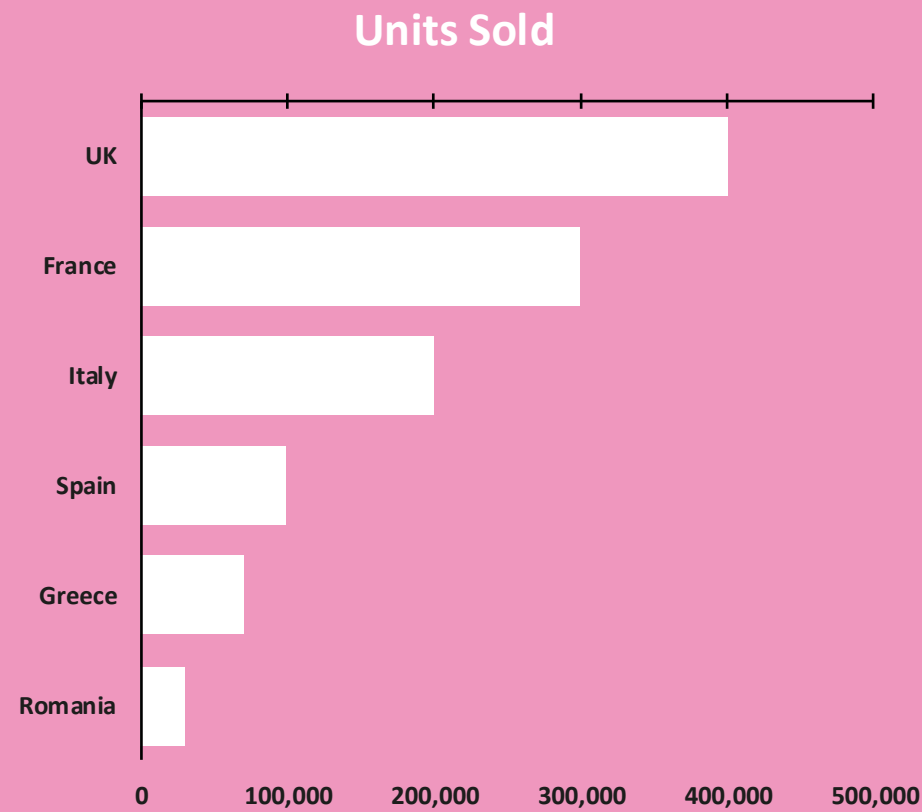
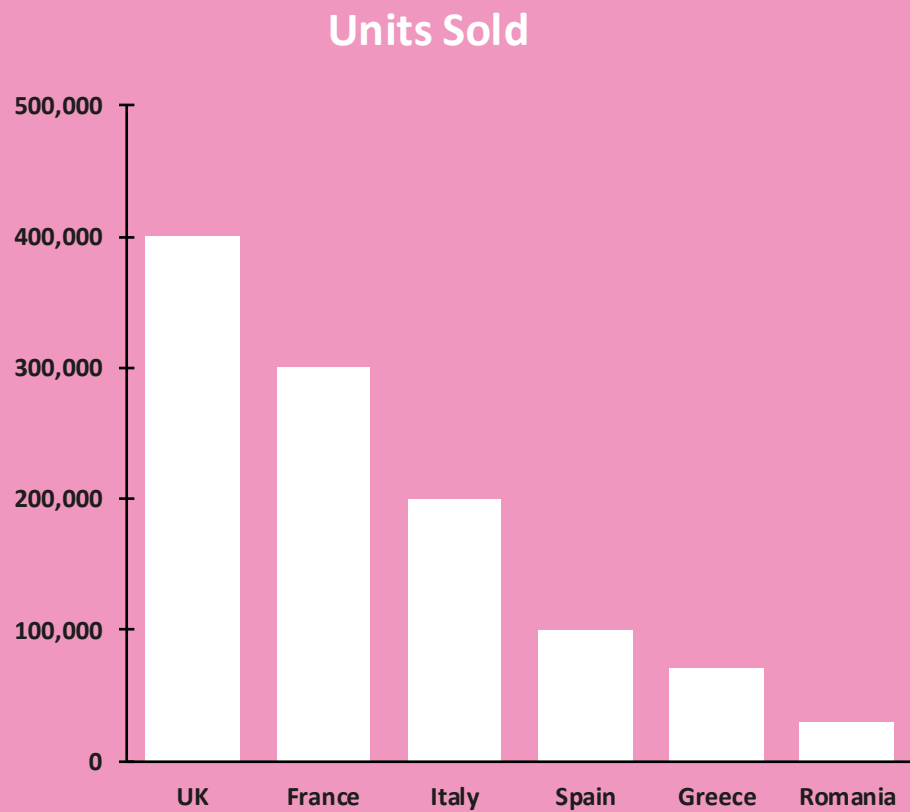
Time Series



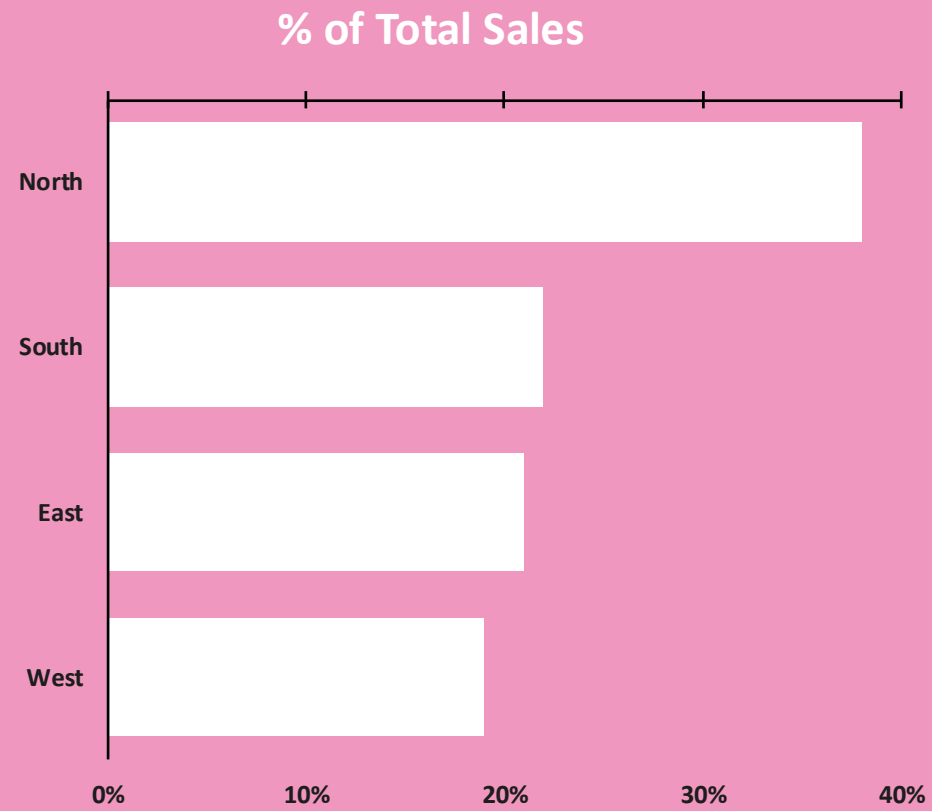
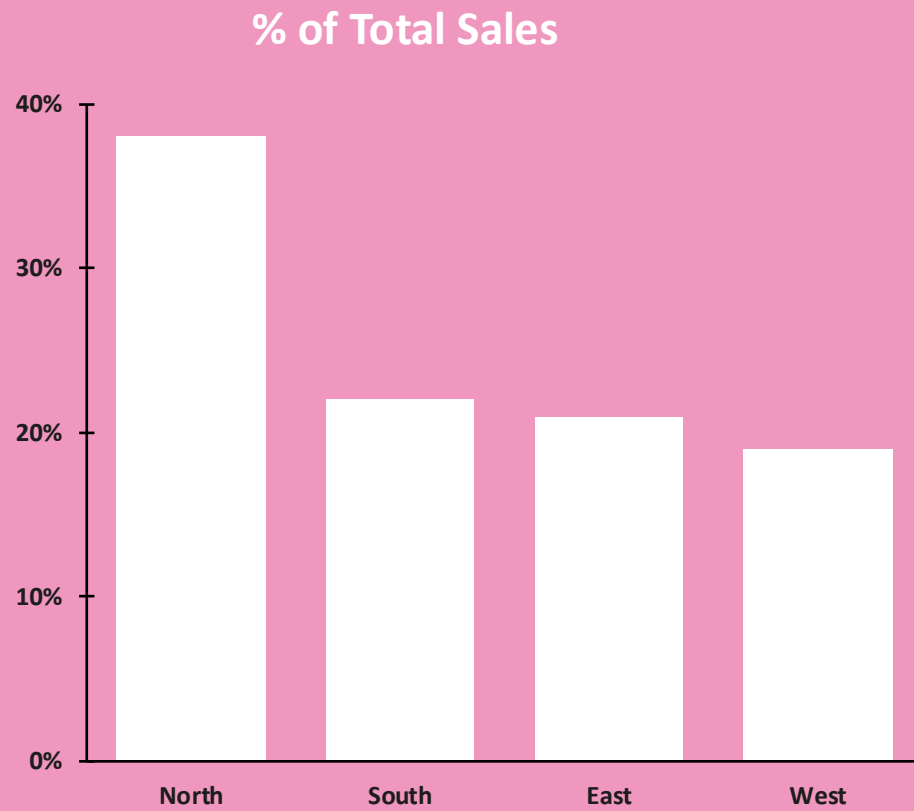
Log Scales



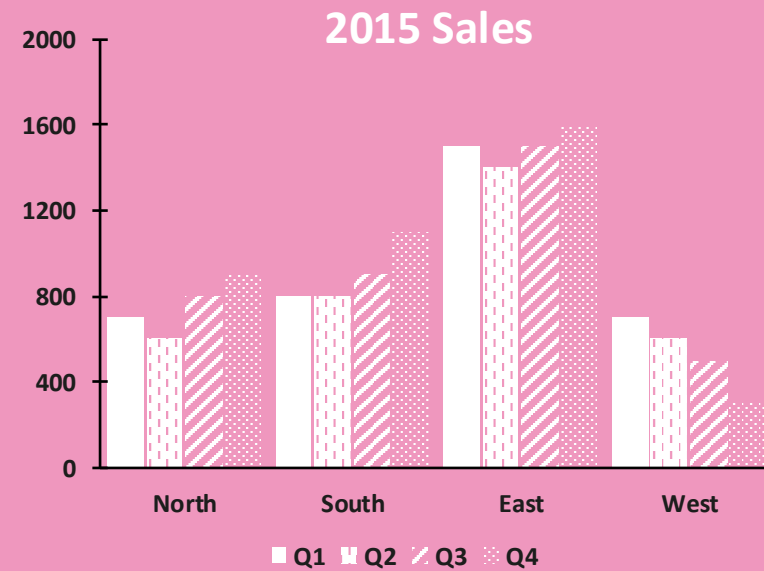
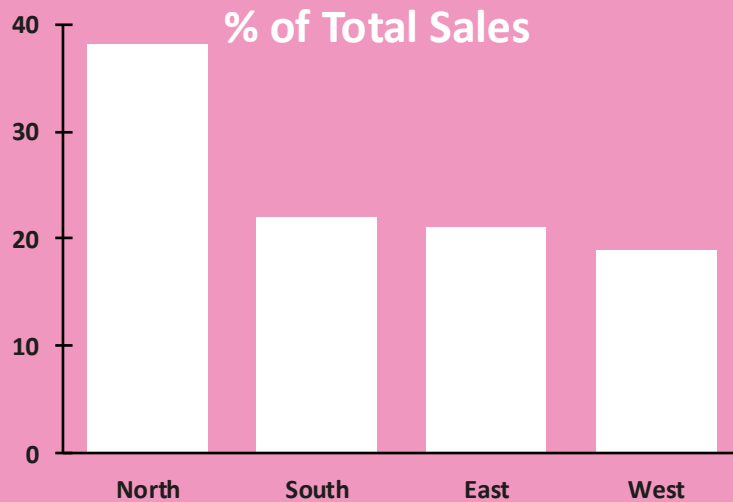
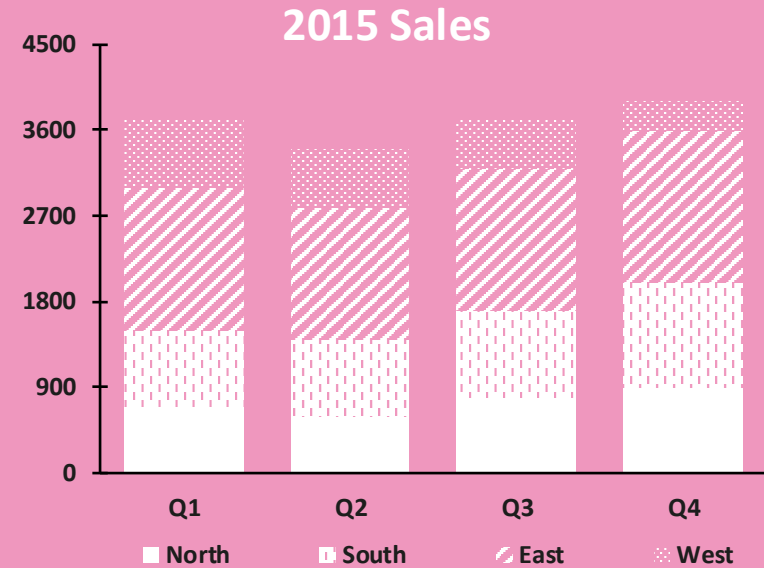
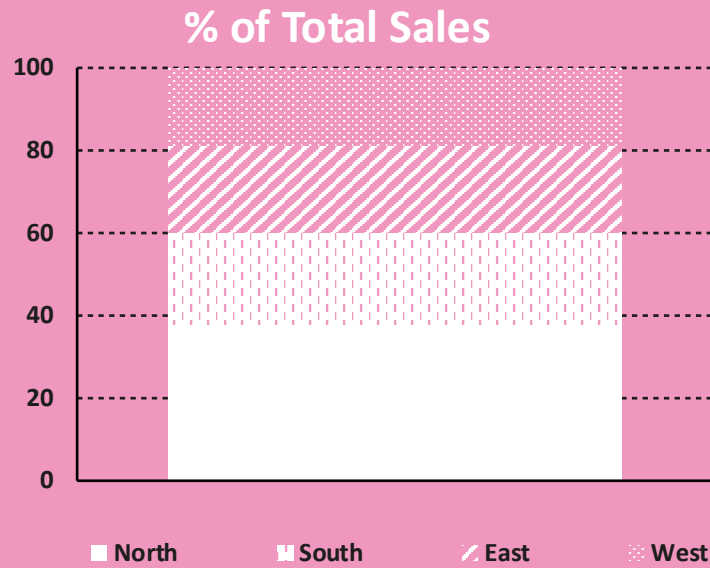
Ranking



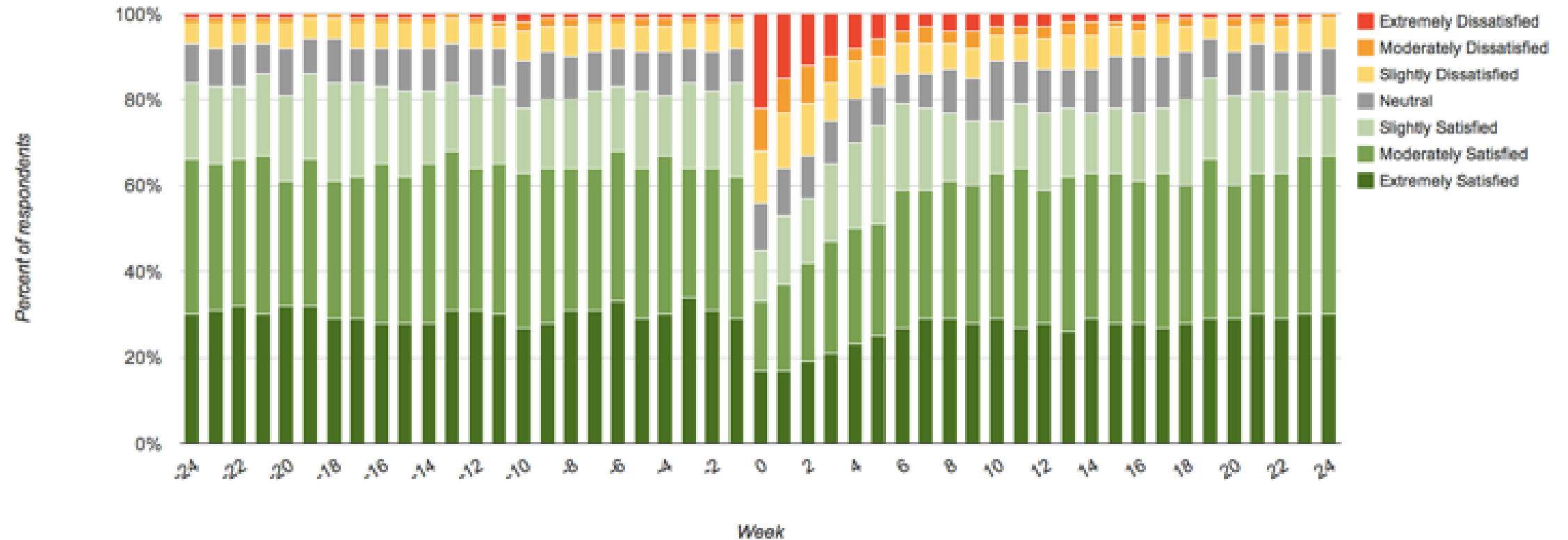
Part-to-Whole



Stacked Bars?



Overall, how satisfied are you with ACME Iron Bird Seed?

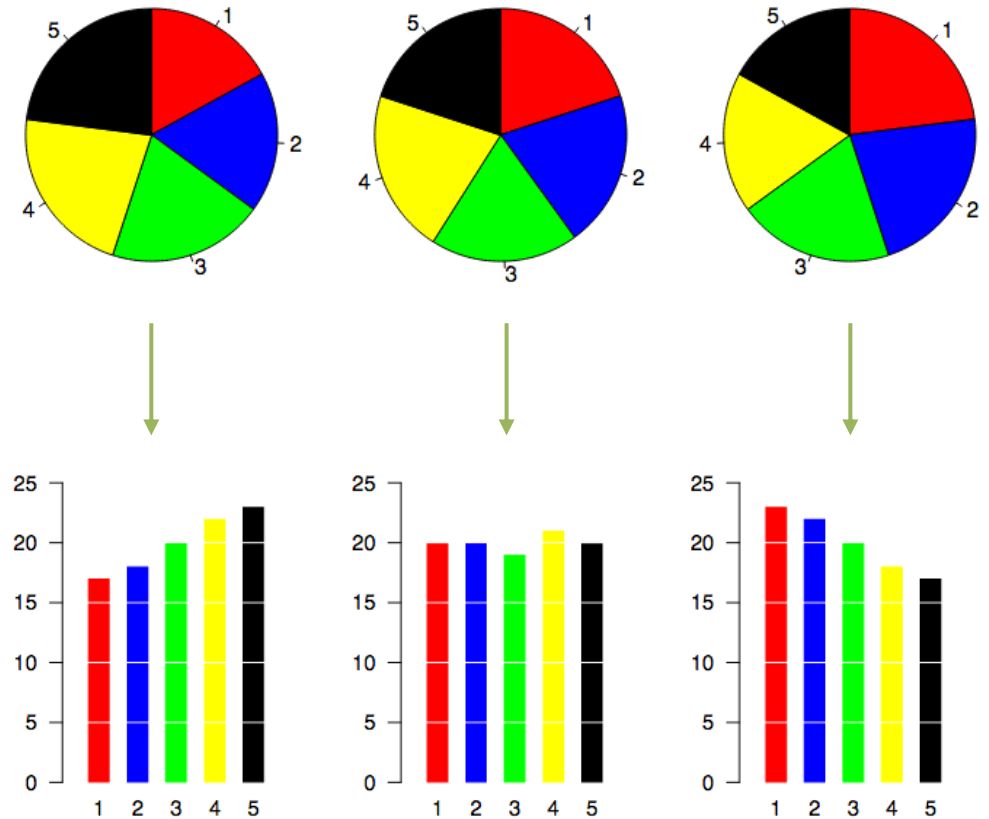


Pie Charts?

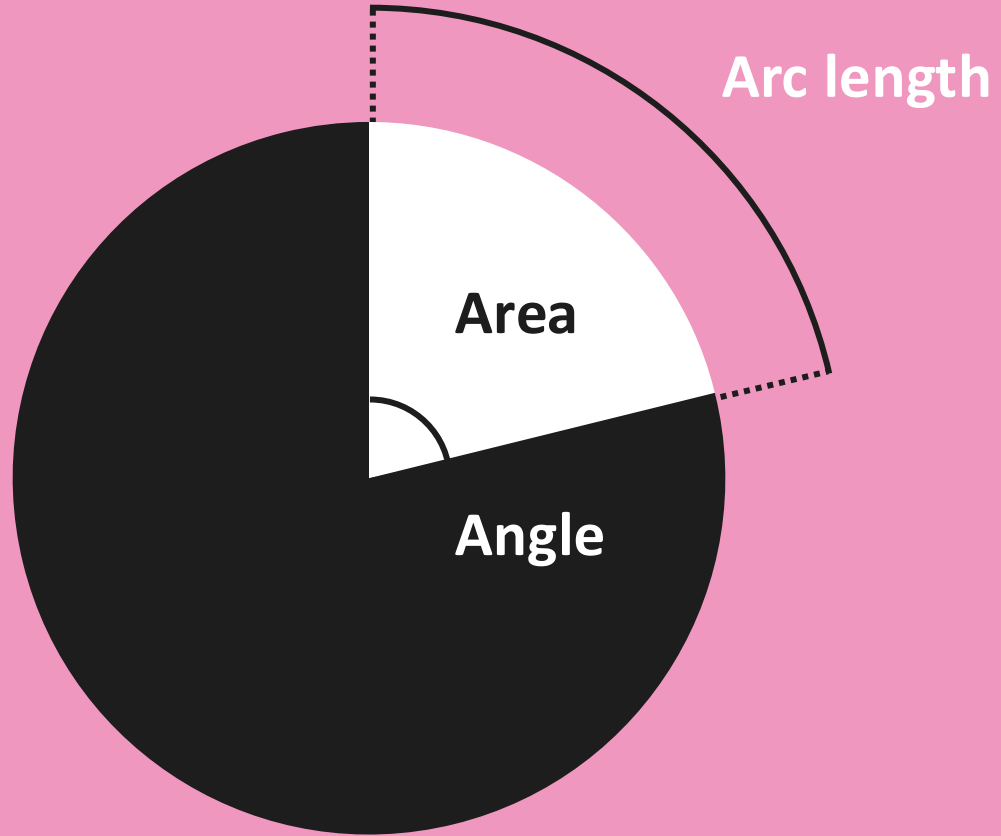
Consider the **purpose** of the visualization

Comparing multiple pie charts is usually **difficult**

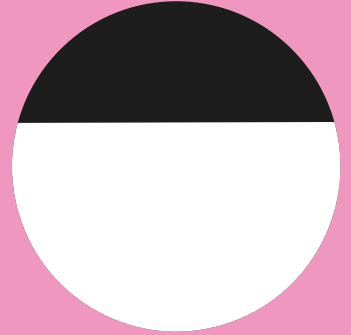
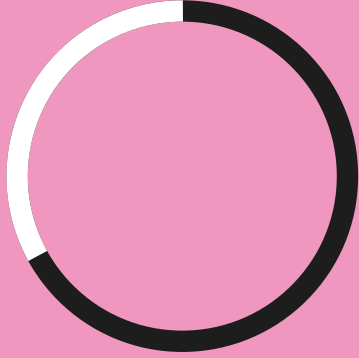
However, if the user goal is to **compare one category vs. the total**, *and* the multiple is **close to 25% or 50%**, then a pie can be more effective than a bar graph.



Pie Charts?

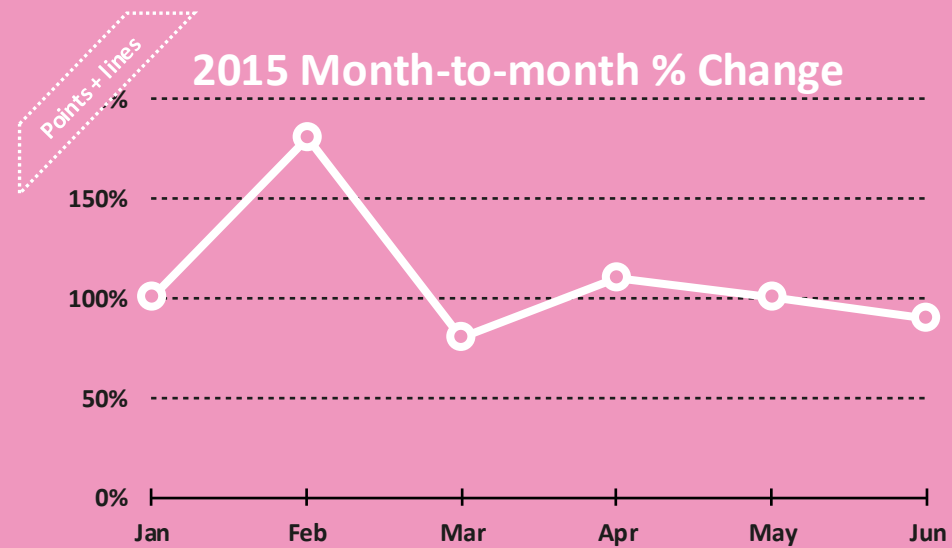
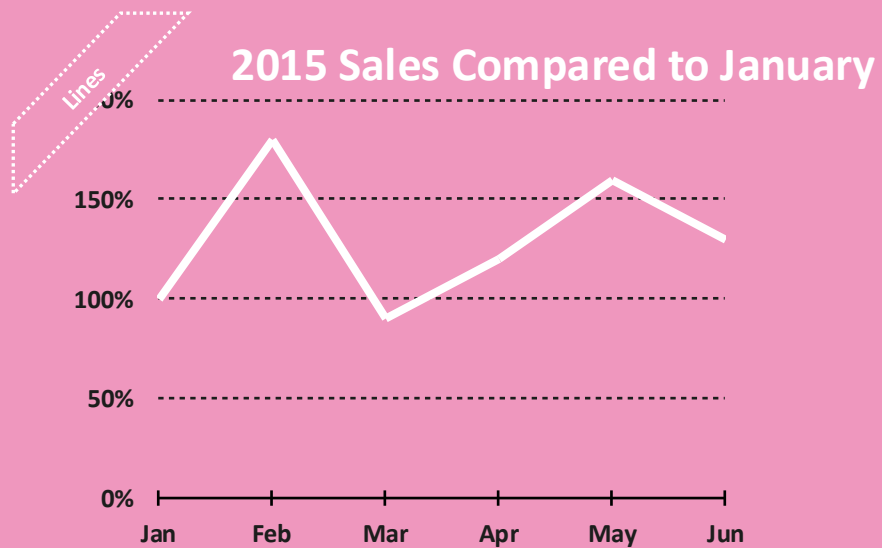
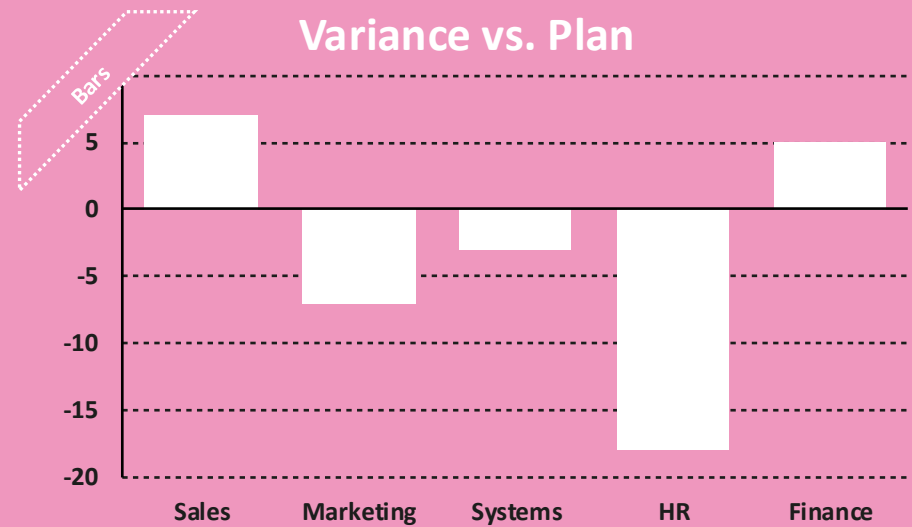
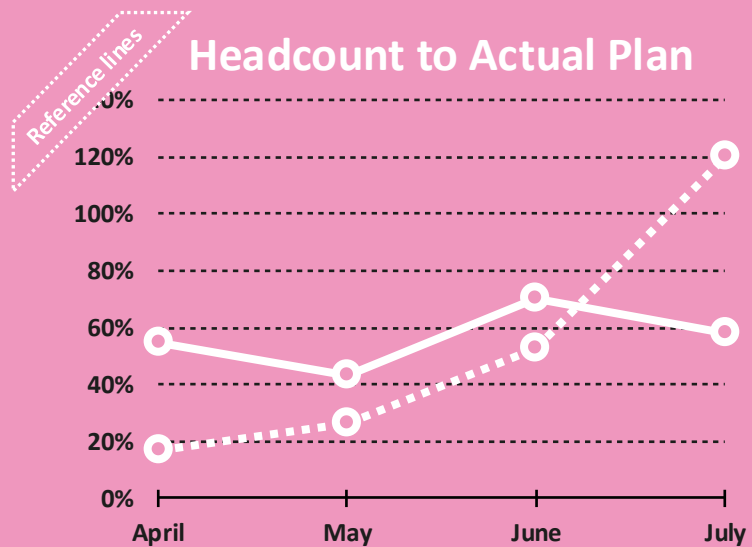


Pie Charts?

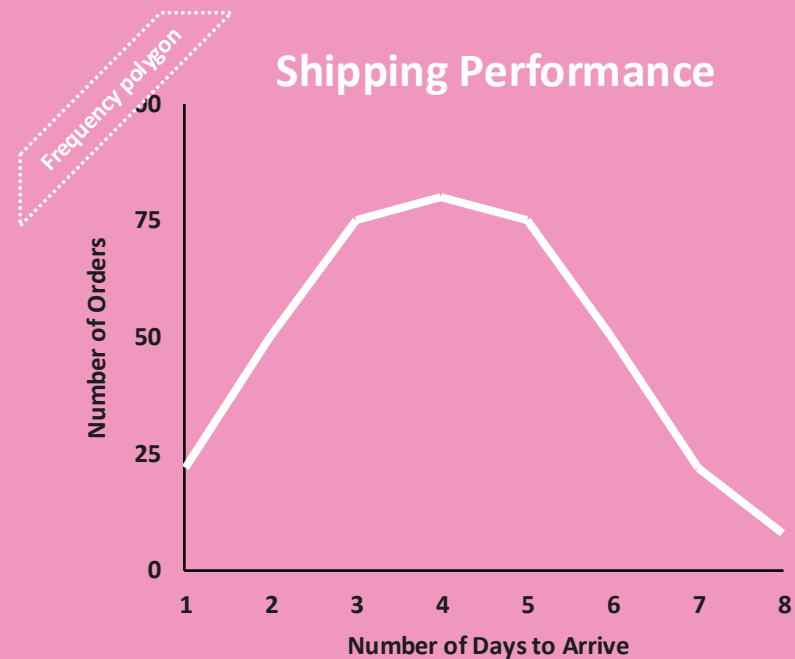
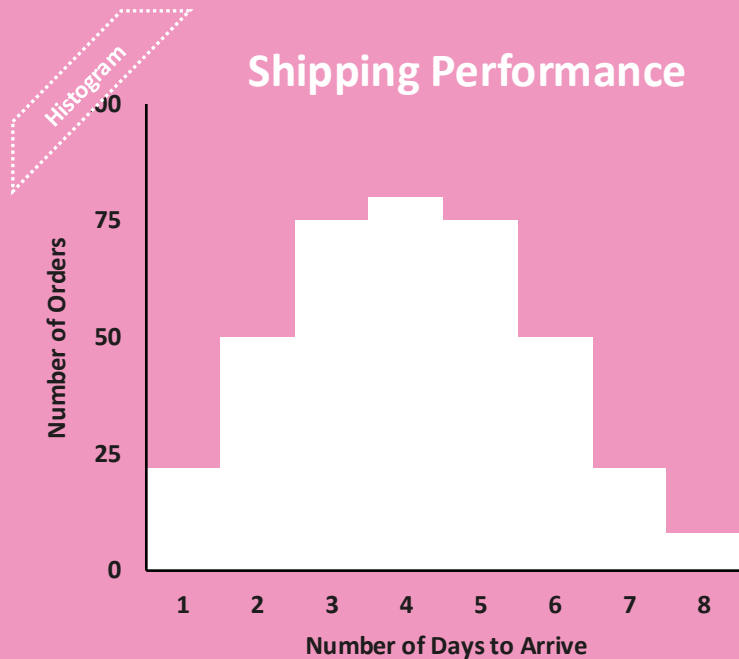


67%

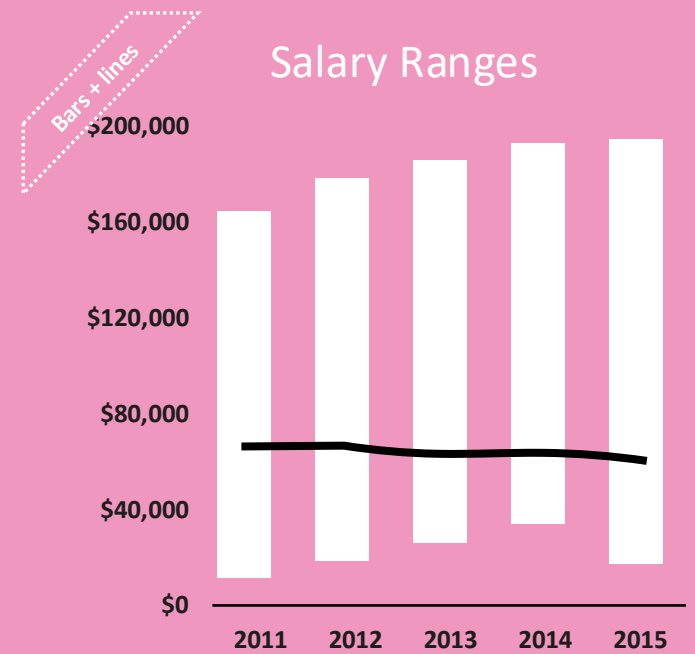
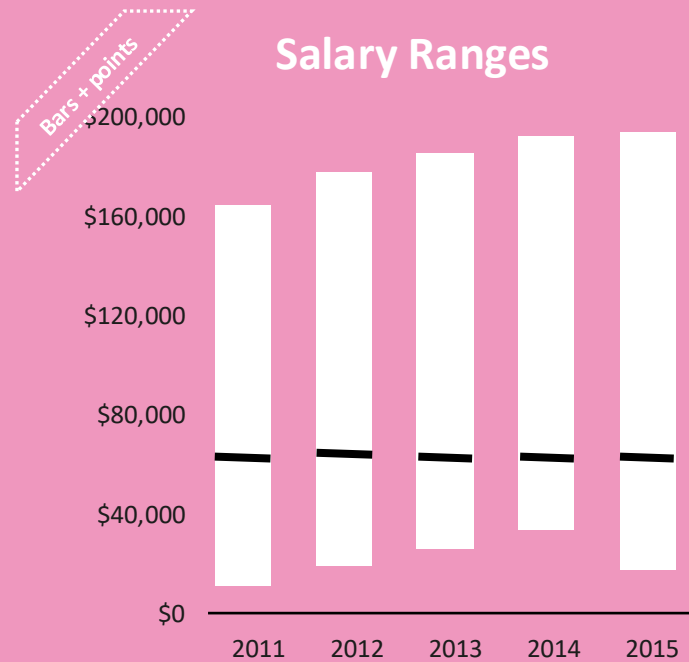
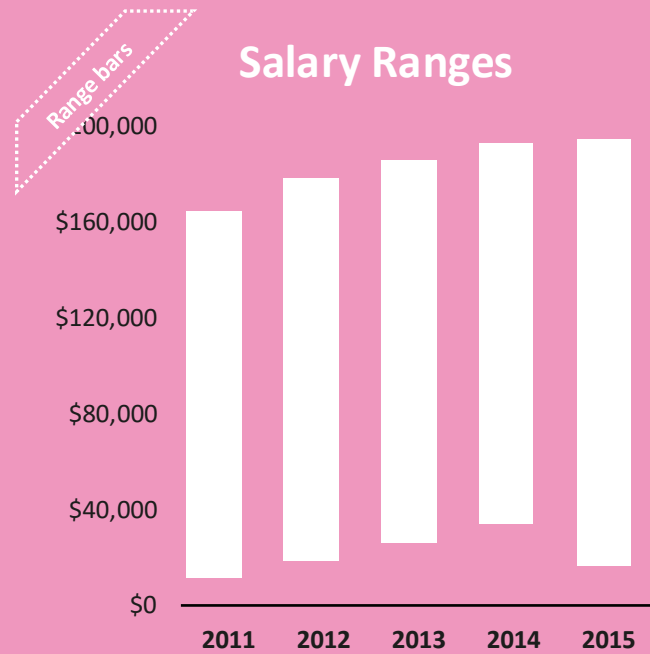
Deviation



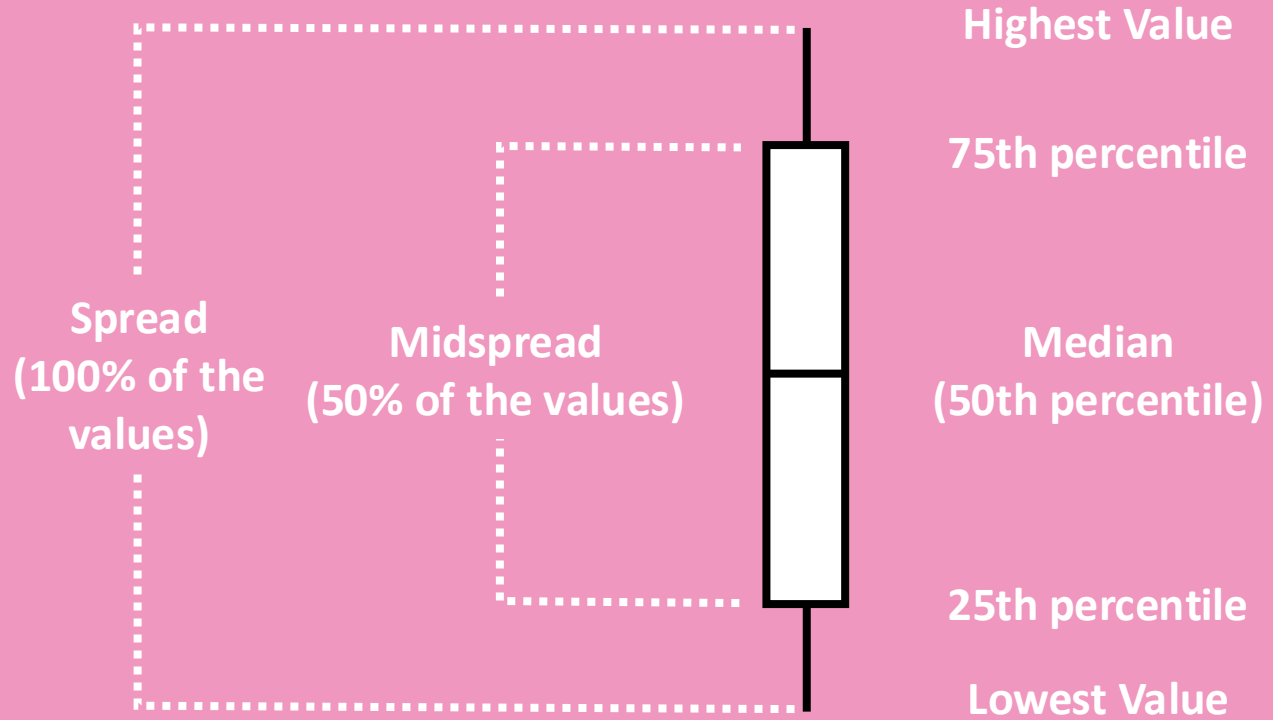
Single Distribution



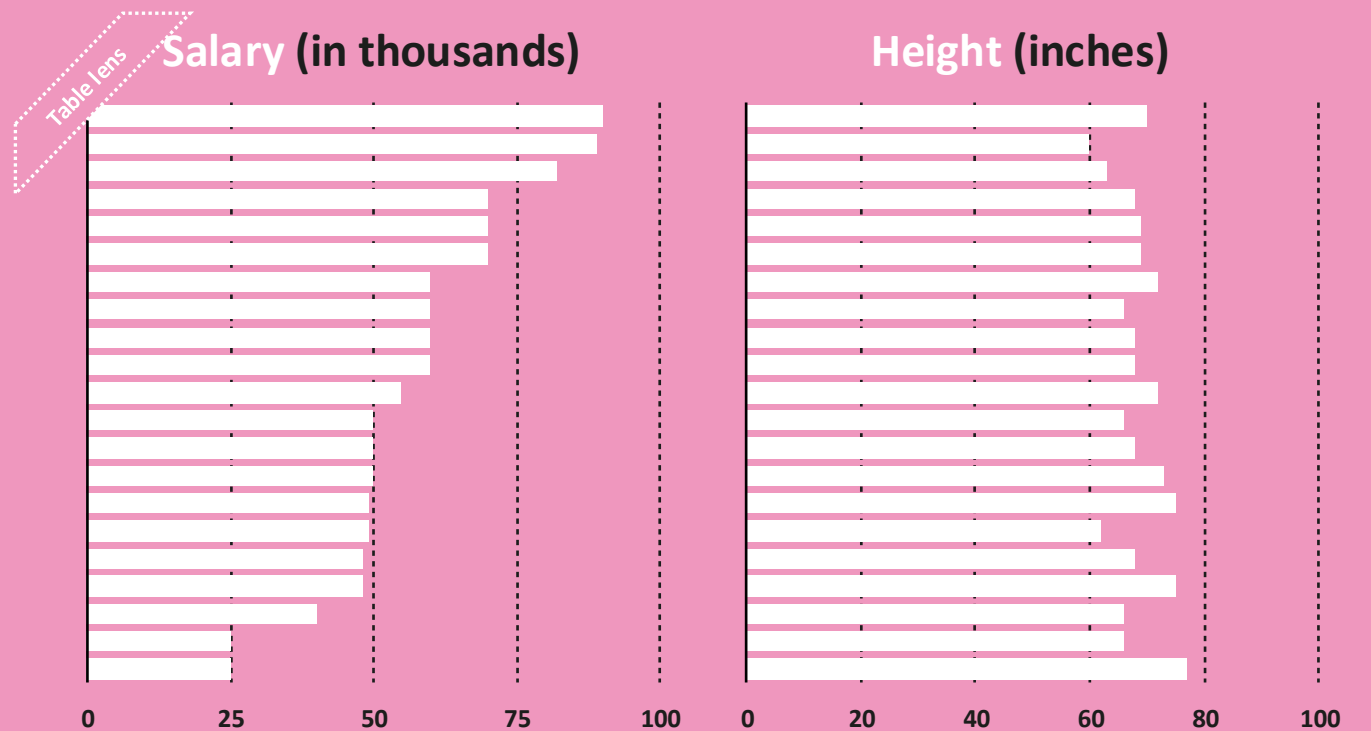
Multiple Distribution



Box Plots

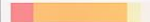
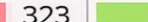
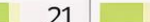
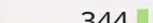
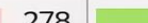
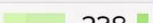
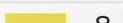
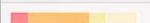
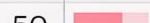
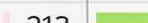
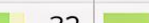
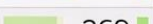
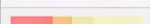
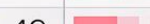
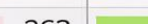
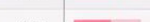
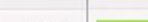
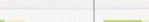
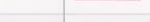
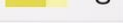
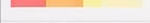
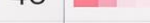
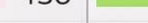
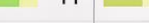
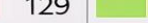
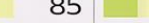
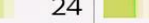
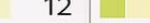
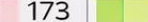
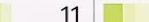
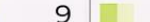
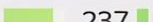
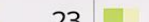
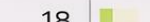
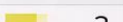
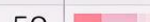
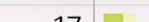
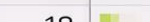
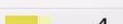
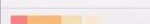
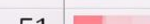
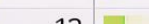
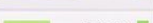
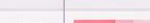
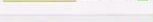
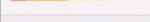
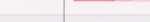
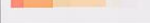
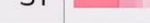


Correlation

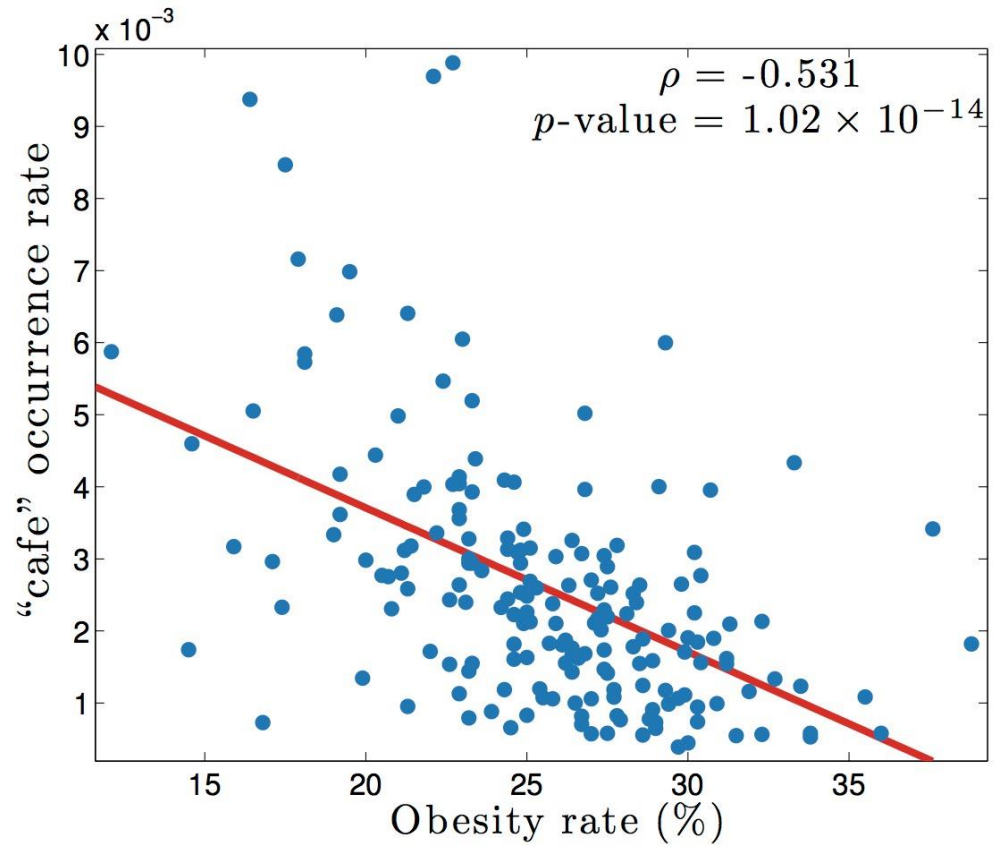
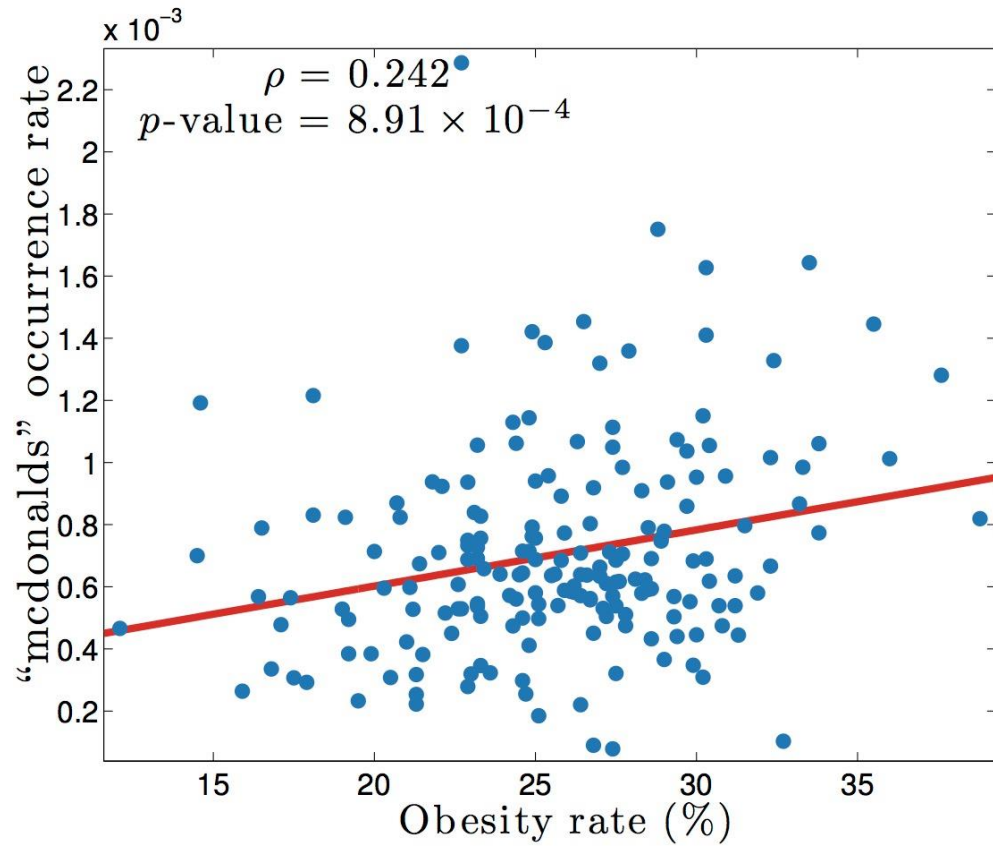


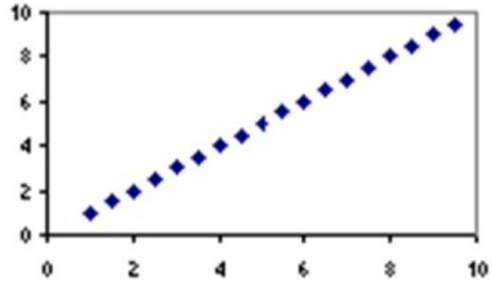
Correlation

Table Lens

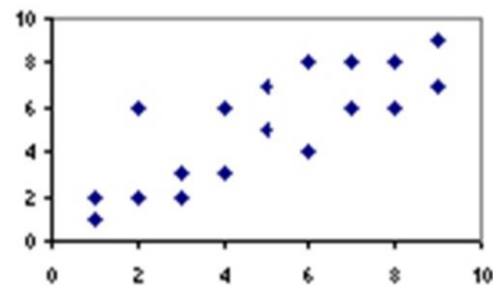
	Name ▾	Calories ⬆	Fat ⬆	Cholesterol ⬆	Sodium ⬆	Carbs ⬆	Carbs ⬆	Protein
357	Chocolate Shake	 357	 10	 50	 312	 63	 63	 1
352	Vanilla Shake	 339	 9	 50	 323	 53	 21	 3
367	French Fries	 344	 9	 55	 278	 63	 21	 2
329	Fish Sandwich	 238	 8	 50	 213	 32	 21	 3
320	Hamburger	 269	 9	 40	 263	 50	 17	 1
329	Ech Dog	 287	 8	 45	 228	 49	 17	 2
322	Scrambled Eggs	 222	 8	 40	 130	 11	 15	 2
343	Egg Custard	 217	 8	 38	 283	 118	 14	 1
355	French Fries	 187	 7	 30	 129	 85	 15	 3
351	Baked Potato	 189	 8	 200	 85	 24	 12	 1
357	Fruit Yogurt	 170	 5	 190	 173	 157	 13	 3
332	Grilled Chicken	 177	 5	 53	 31	 11	 9	 1
352	Roast Chicken	 237	 5	 70	 50	 23	 18	 1
351	Hamburger	 257	 3	 50	 25	 17	 18	 1
309	Fish Sandwich	 226	 4	 51	 112	 12	 8	 1
357	Chicken Nuggets	 283	 3	 56	 85	 27	 7	 1
110	Total Cereal	 210	 2	 31	 85	 10	 6	 3
110	Wheaties	 203	 0	 0	 30	 0	 0	 1
110	Frosted Flakes	 110	 0	 0	 10	 0	 0	 3

Patterns: Direction

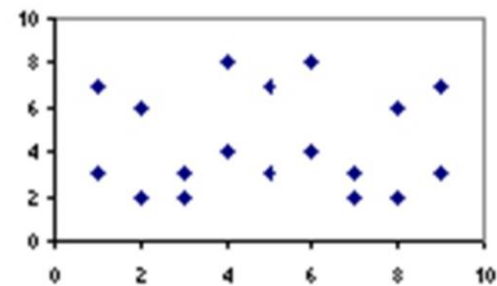




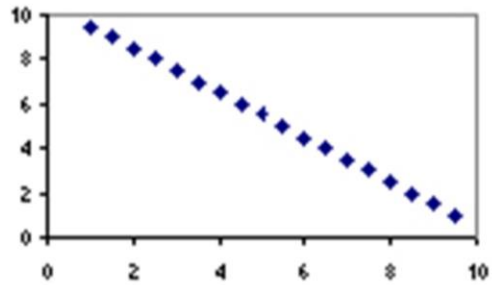
Maximum positive correlation
($r = 1.0$)



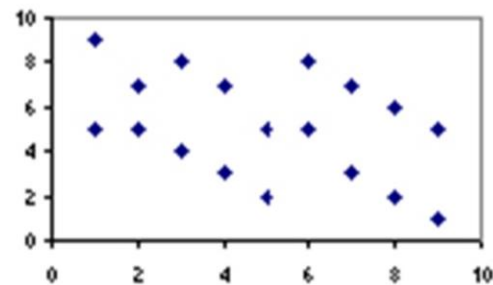
Strong positive correlation
($r = 0.80$)



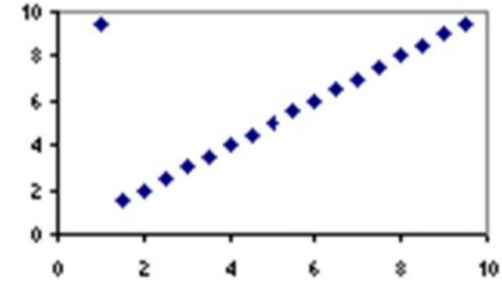
Zero correlation
($r = 0$)



Maximum negative correlation
($r = -1.0$)

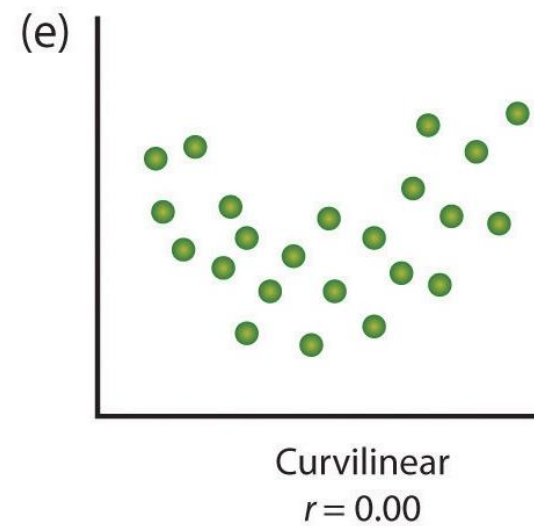
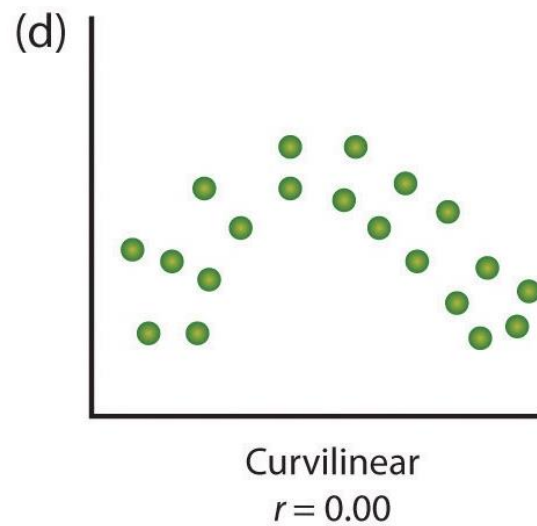
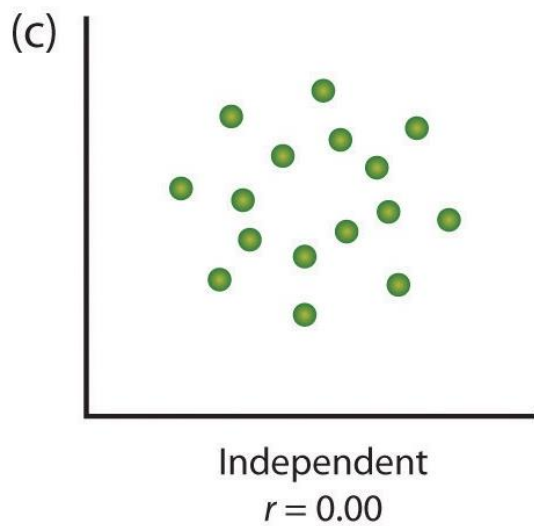
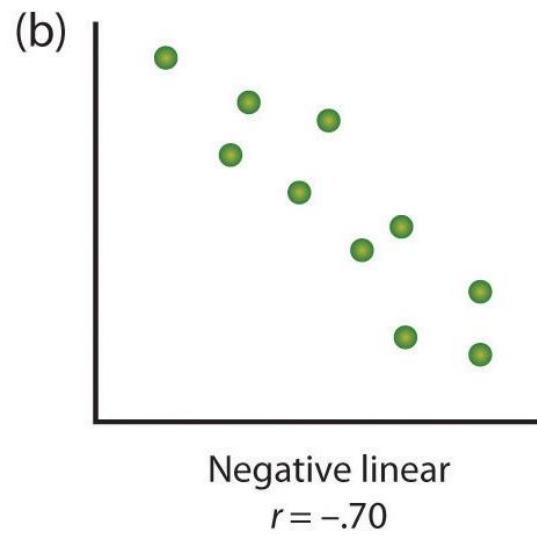
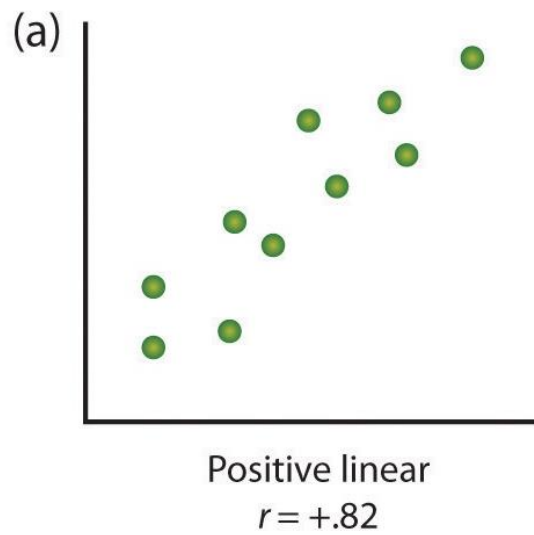


Moderate negative correlation
($r = -0.43$)

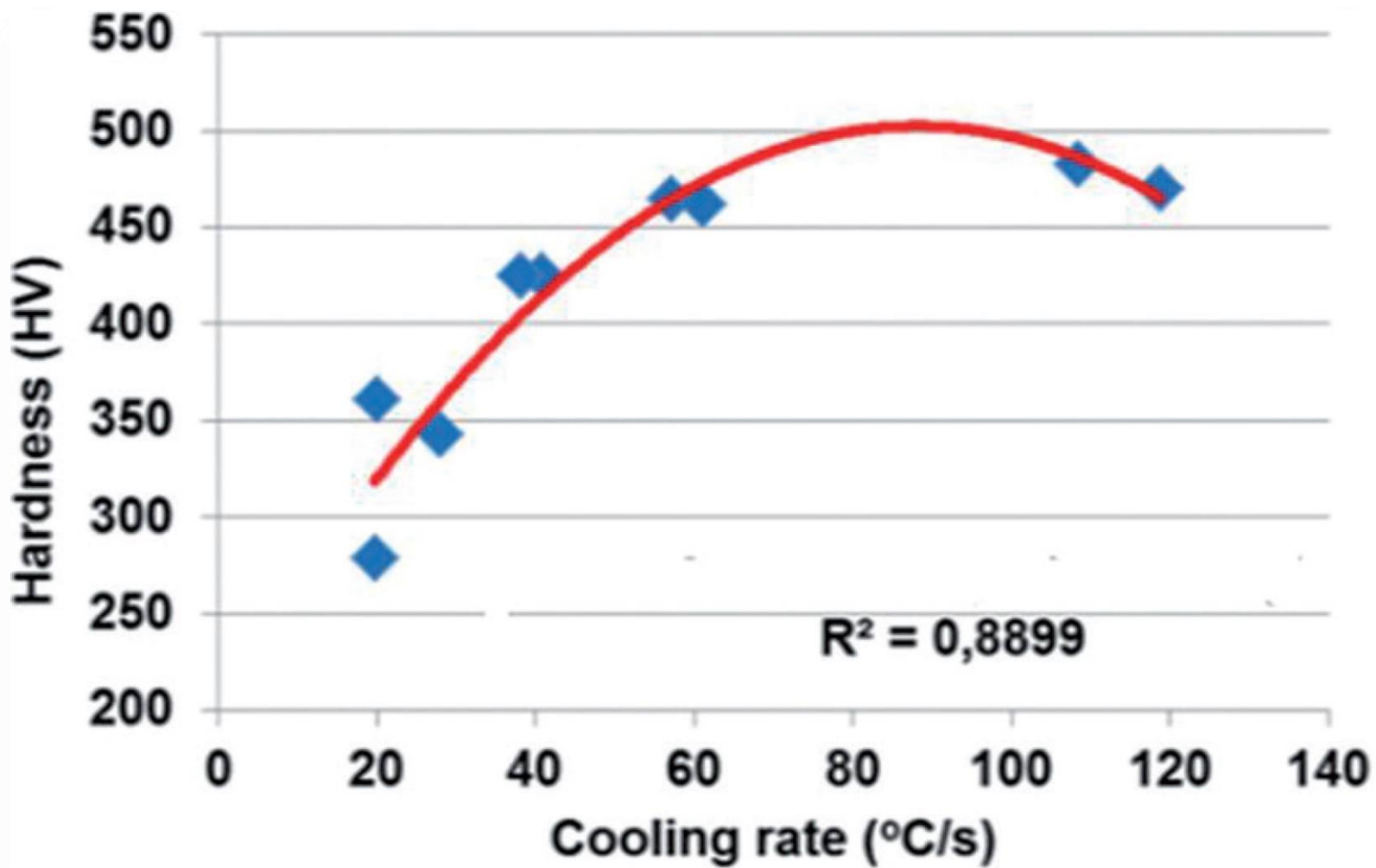


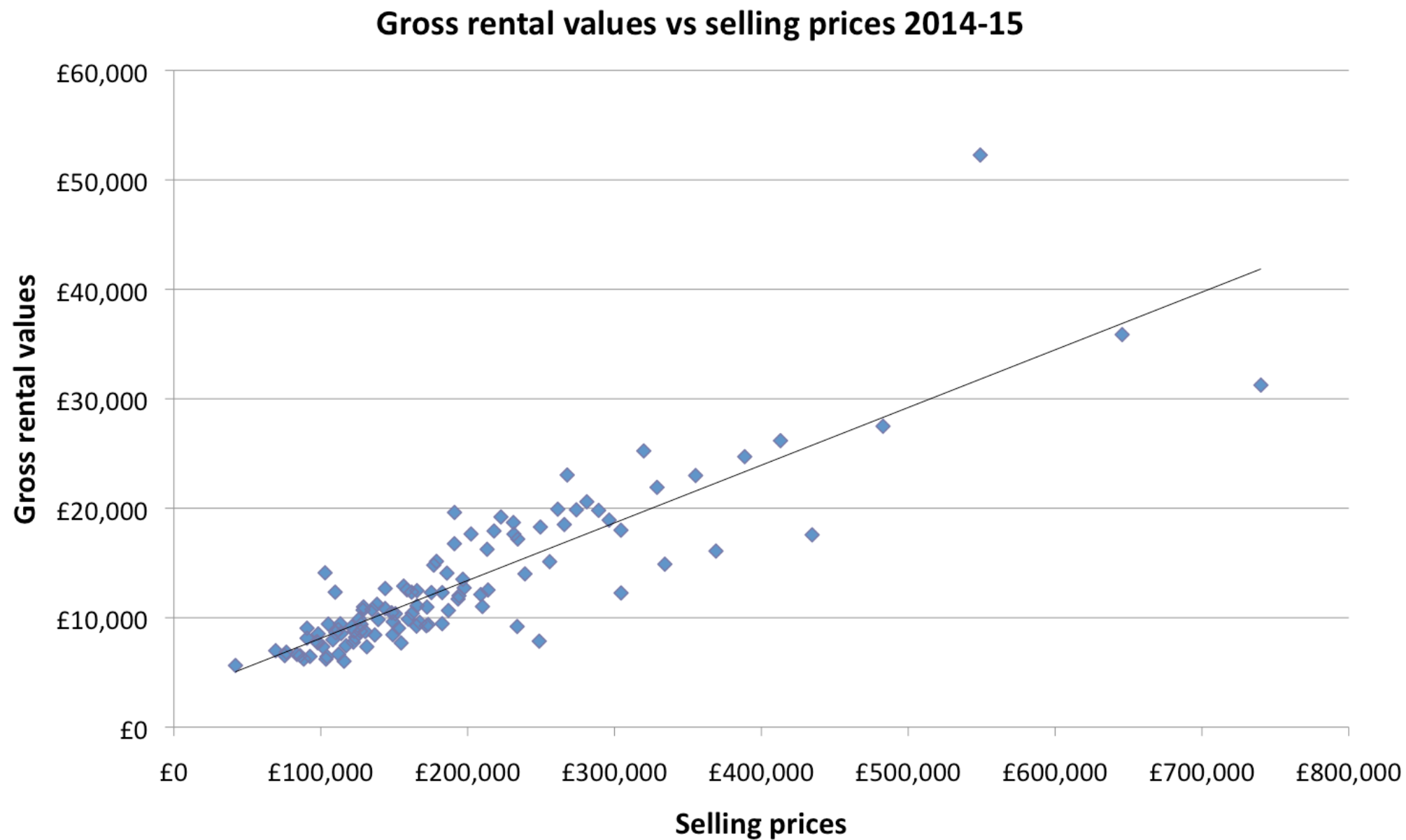
Strong correlation & outlier
($r = 0.71$)

Patterns: Shape

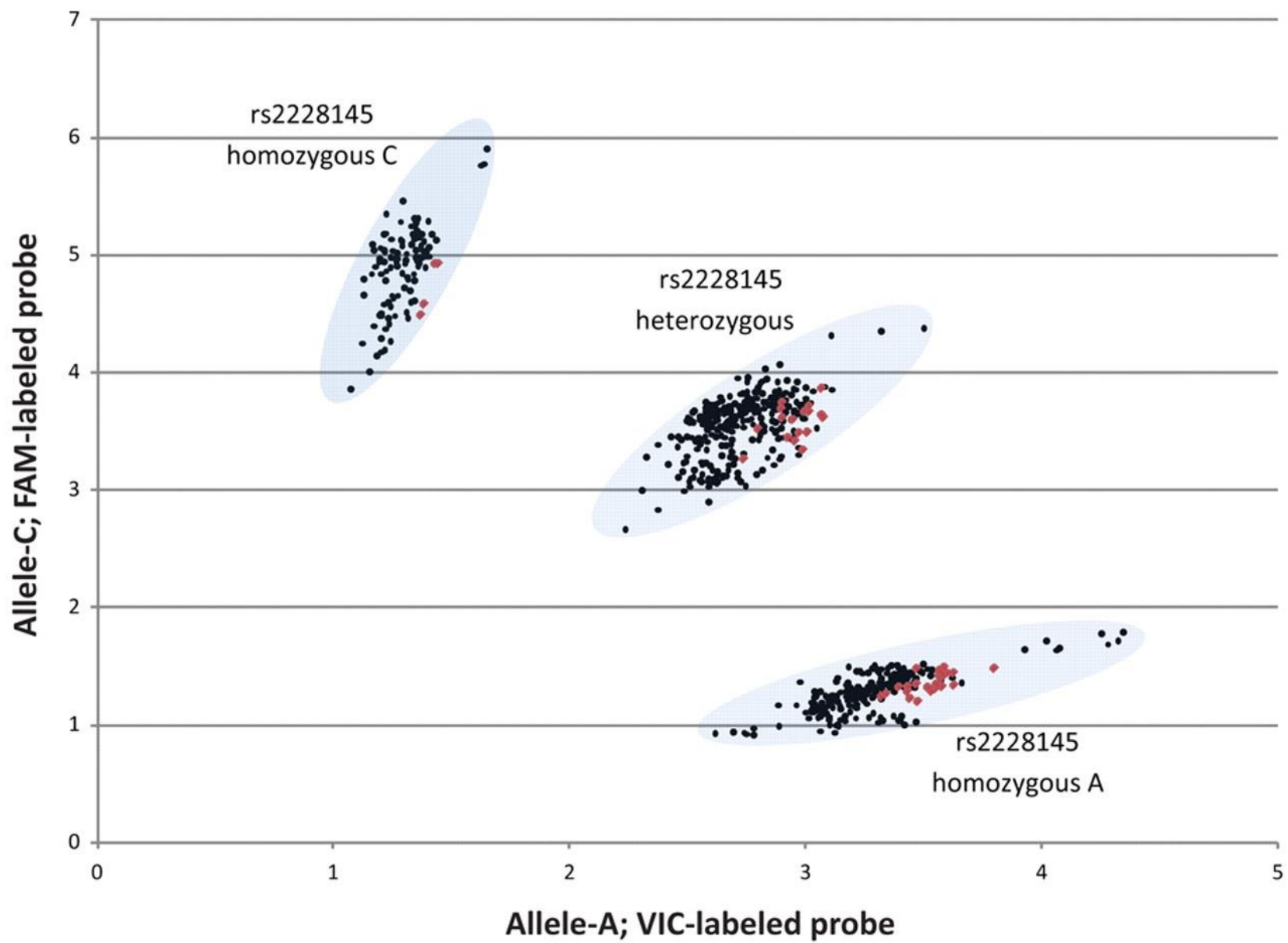


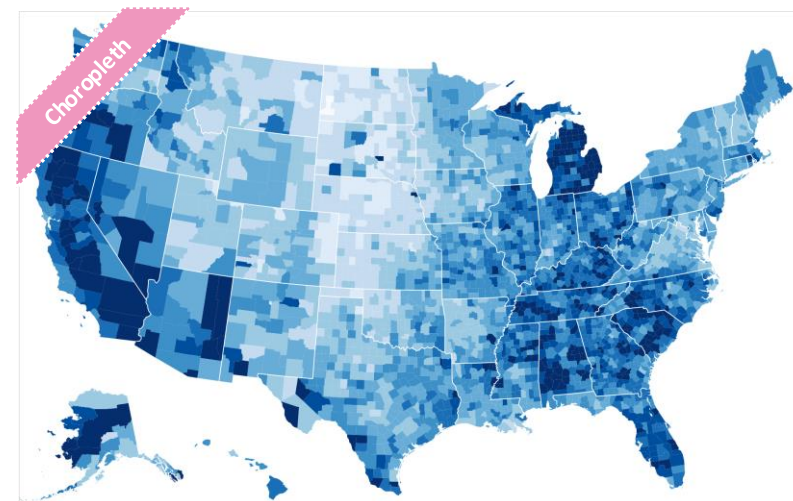
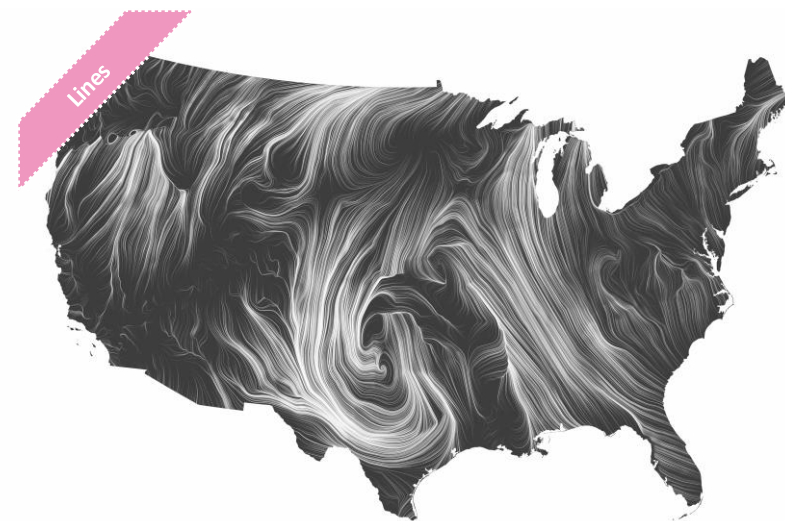
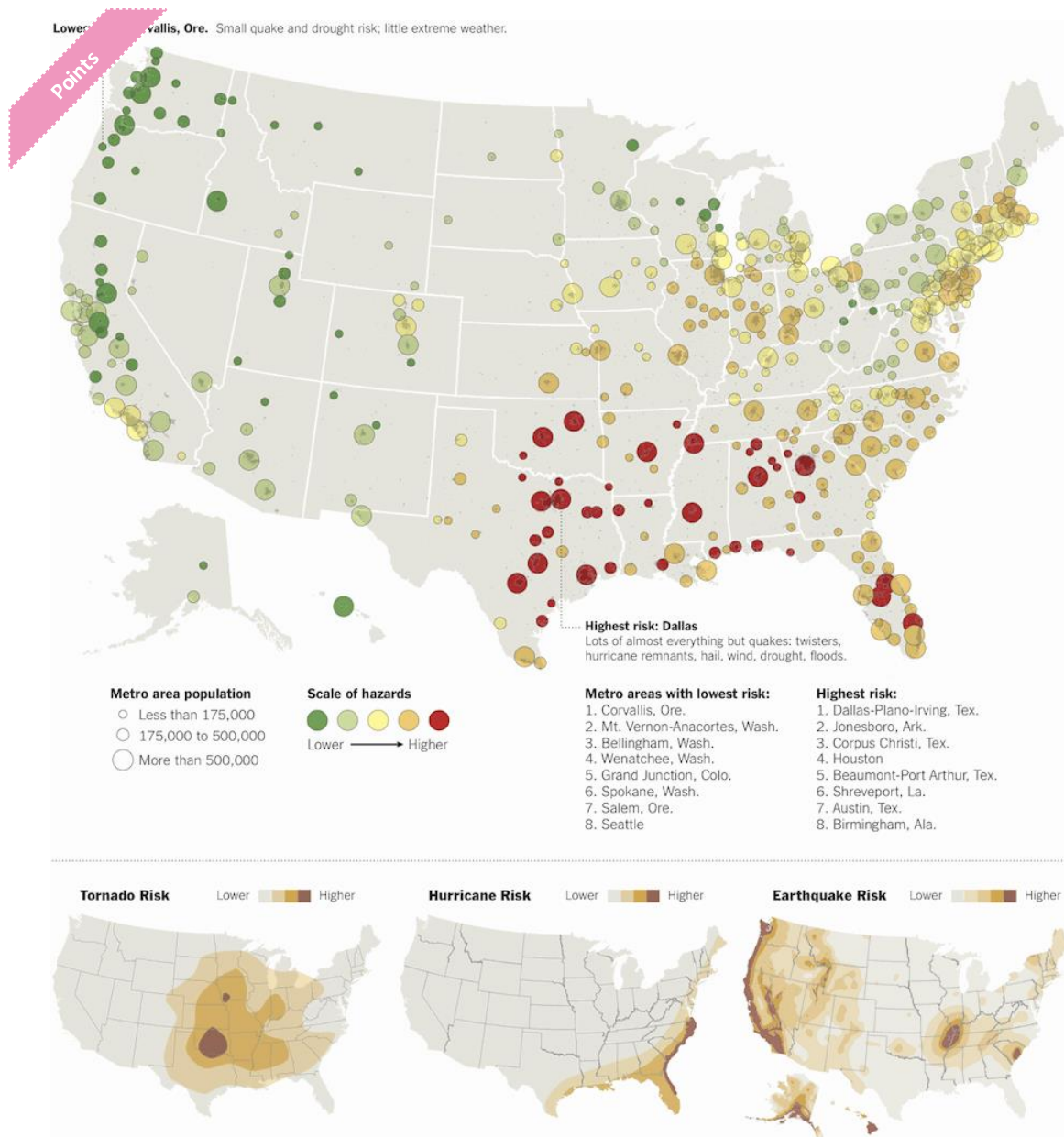
Patterns: Gaps



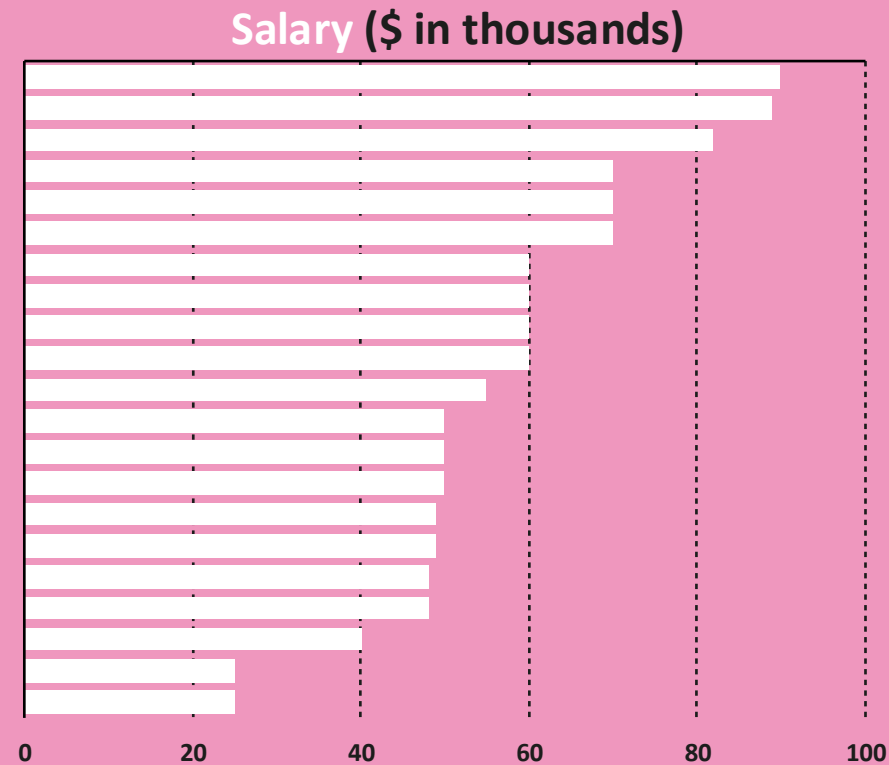
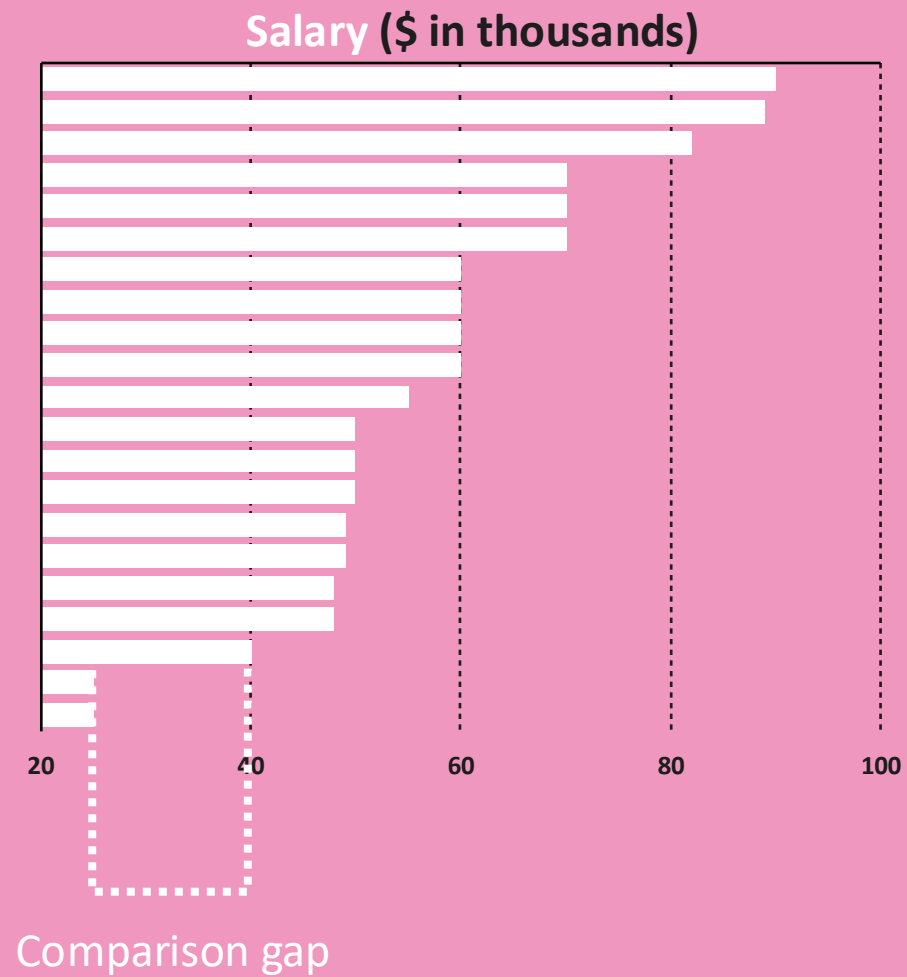


Concentrations





What's wrong here?



Map key:

Cities visited

1979-1980

1981-1985

1986-1990

1991-1995

1996-2000

2001-2004

200 —

150 —

100 —

50 —

0

**Total number of
cities visited**

1979-80

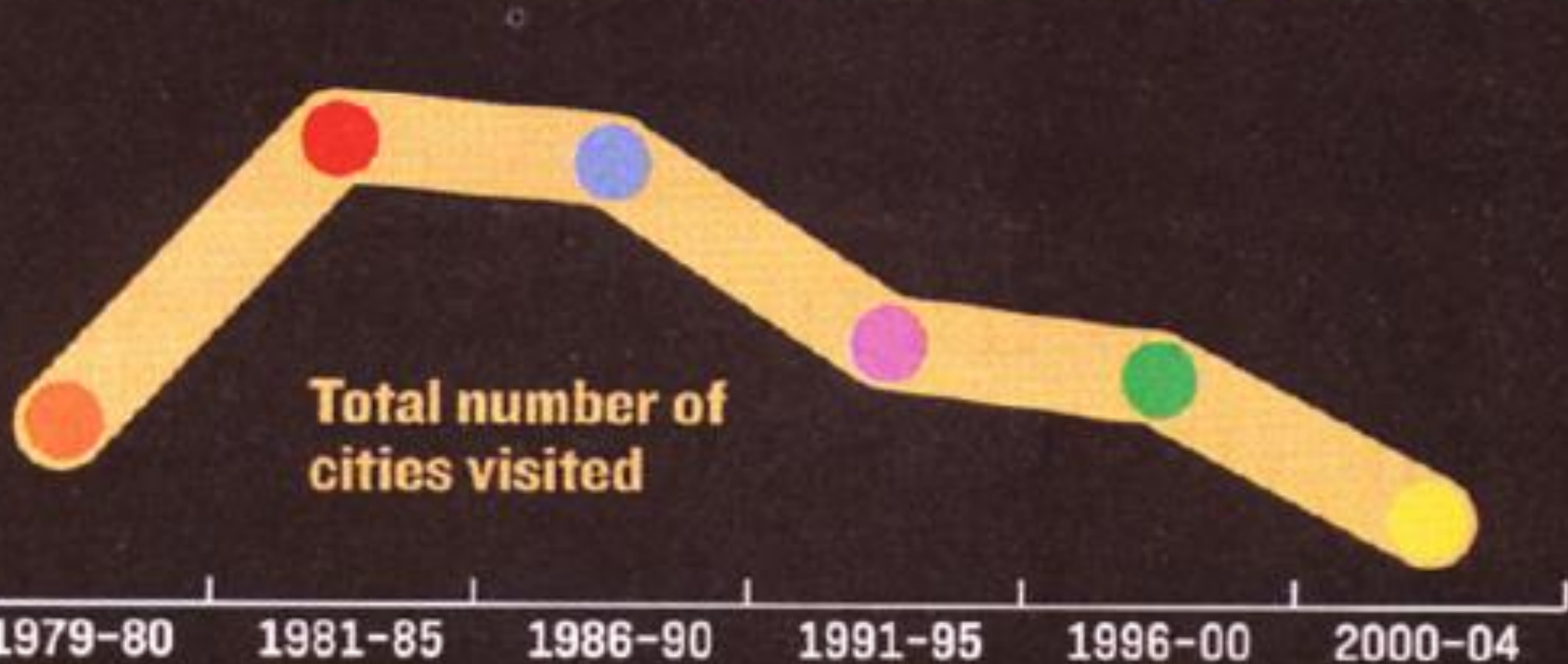
1981-85

1986-90

1991-95

1996-00

2000-04



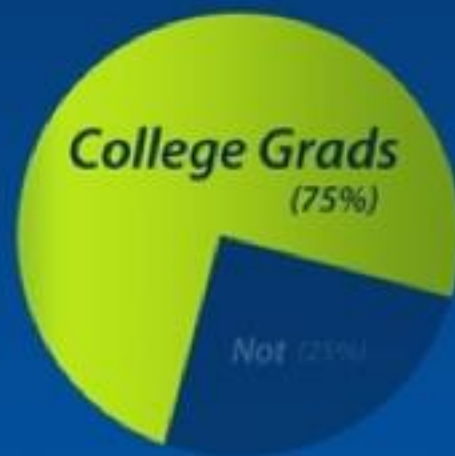
COMMUTING TO WORK

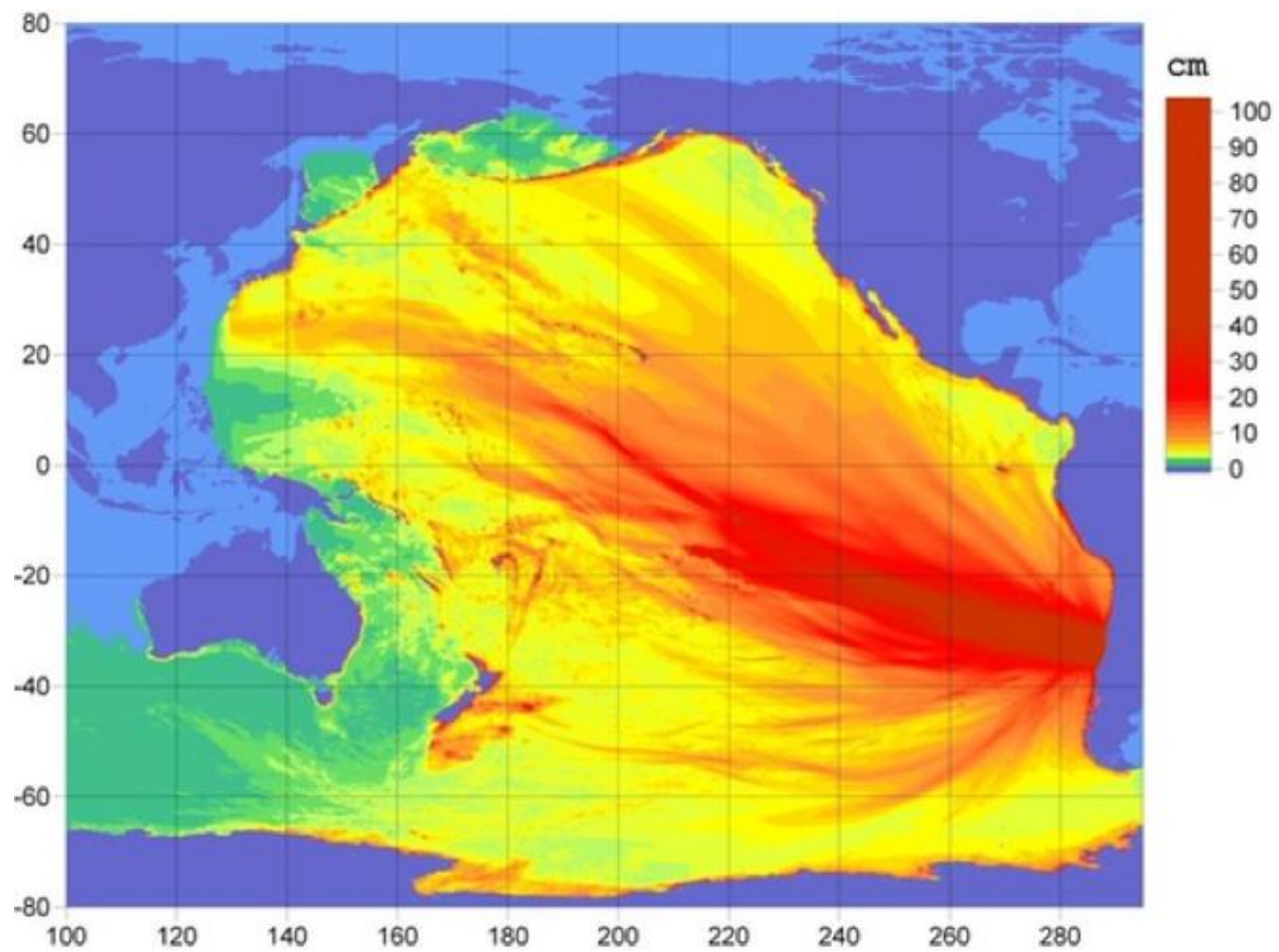
CHICAGO / LOS ANGELES / NEW YORK CITY / ATLANTA
SAN FRANCISCO / HOUSTON / WASHINGTON / SEATTLE



Who's Blogging Right Now?

(% of the blogosphere)

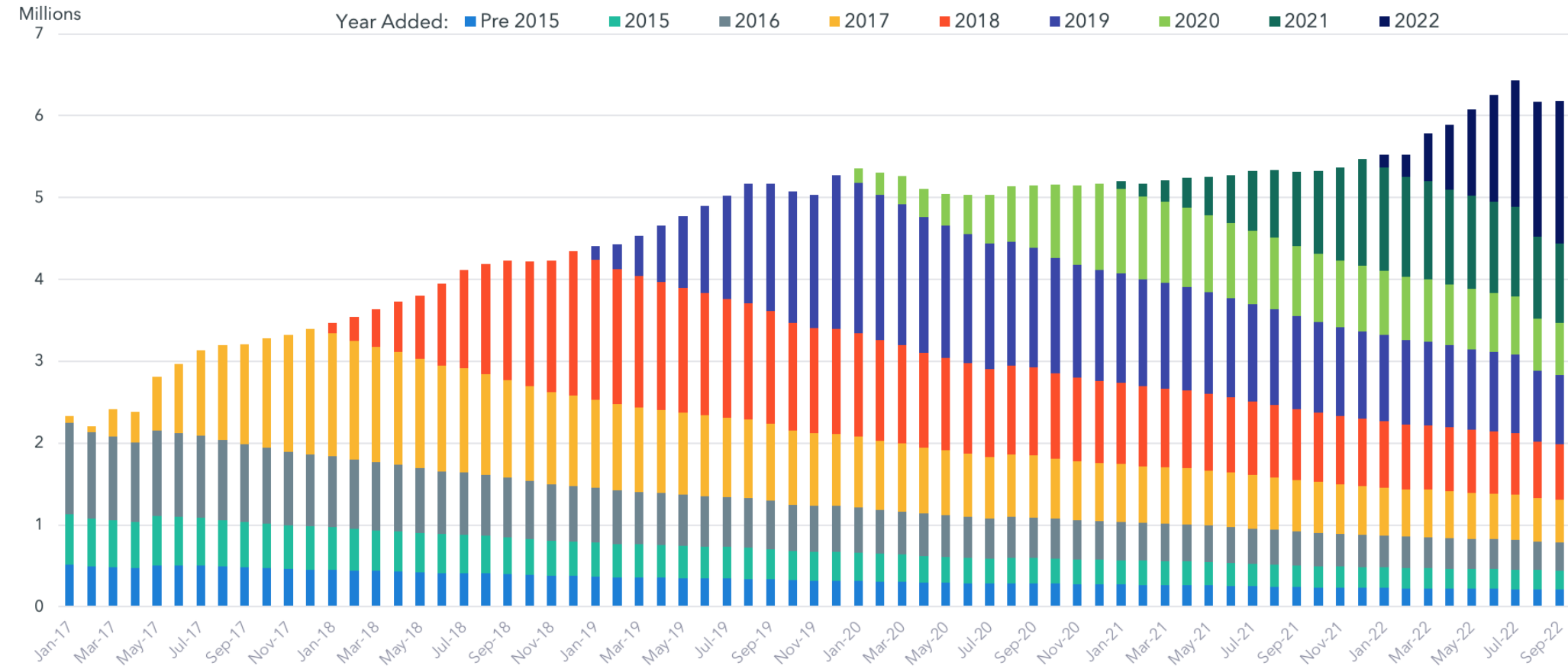




54% of Airbnb Listings have been Added since 2020

AIRDNA

Airbnb: Global Active Listings by Year Added



Source: AirDNA

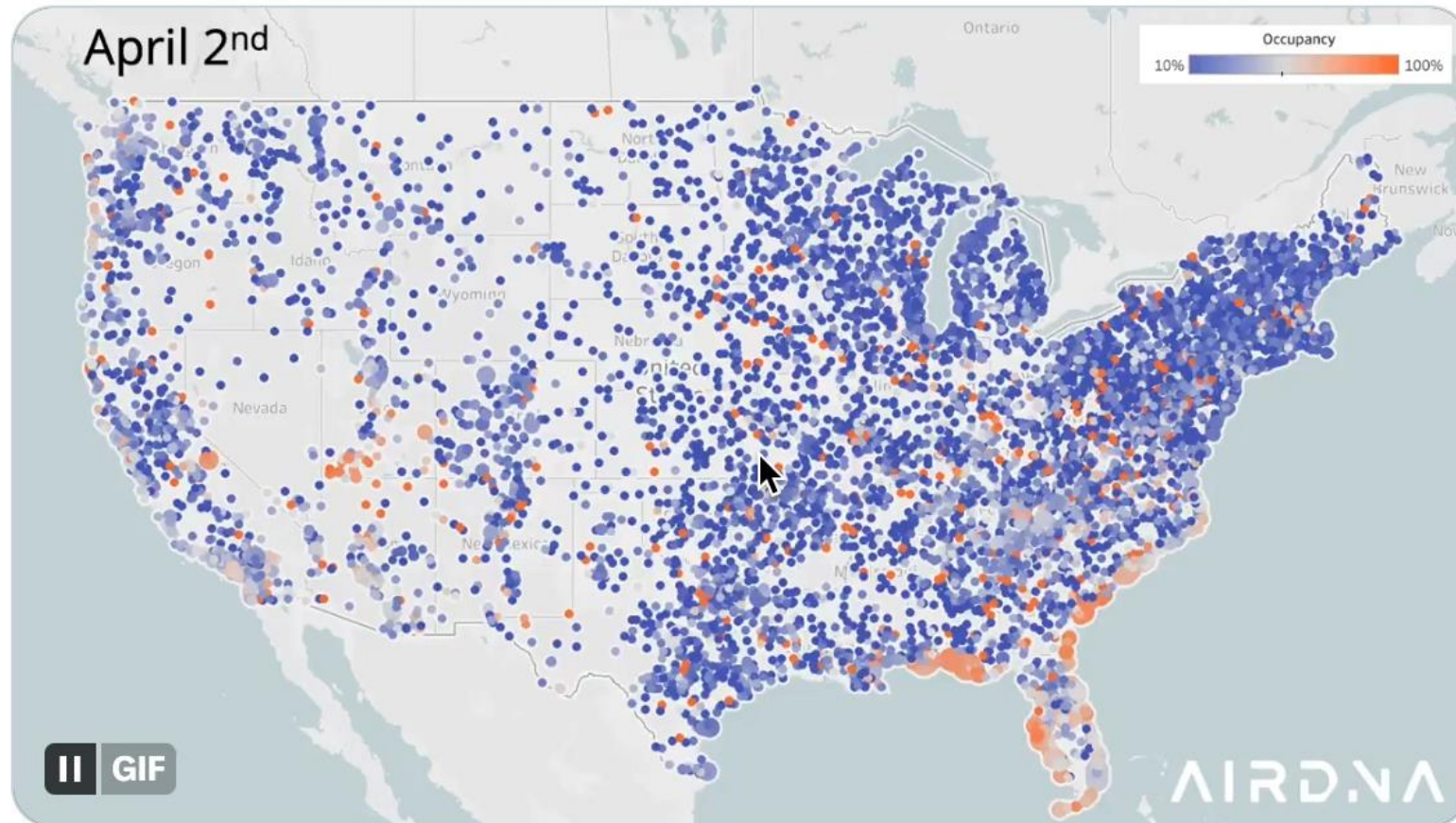


Jamie Lane ✓

@Jamie_Lane



With the eclipse approaching in less than a week, over half of U.S. cities along the eclipse's path are fully booked for the night of April 7th according to data from [@airdna](#)



2:58 PM · Apr 2, 2024 · **659.8K** Views

Finding
Data

**There is an extensive
list of potential data
sources in the
resources section on
canvas!**

US National Data Sources

Wikipedia list of [National and International Statistical Services](#)

[data.gov](#), [data.seattle.gov](#)

US Census is a good source for equity/inequity related information.

International Services

[World Health Organization Global Health Observatory Repository](#)

[United Nations Data Portal and API](#)

[World Bank](#)

[European Union National government data sources](#)

Other Sources

Regulatory organizations that deal with an area of interest (e.g., Centers for Disease Control)

Professional Bodies/Unions (e.g., [American College of Surgeons Cancer Database](#))

Charitable organizations and nonprofits often focus on specific health issues [like Cancer](#)

Companies themselves:

- Annual reports

- Financial reports

[Open Corporates](#) - public information about companies

[Investigative Dashboard](#) - very large list of international datasets and data holders

[Guardian Data Store](#) - all of their datasets

[datacatalogs.org](#)

[USAFacts.org](#)

Public Application Programming Interfaces (APIs):

Sound Transit API

twitter

Facebook

Open Data Institute lists other APIs

“Screen scraping/Web scraping”

Google/Bing/DuckDuckGo/etc. it!

Break

until x:xx pm

Data Formats

Data Formats

Plain Text

Comma Separated Values (CSV)

XML (eXtensible Markup Language)

JavaScript Object Notation (JSON)

```
1 Johnson, Smith, and Jones Co. 345.33 Pays on time
2 Sam "Mad Dog" Smith          993.44
3 Barney & Company              0      Great to work
always pays with cash.
4 Johnson's Automotive          2344
```

Data Formats

Plain Text

Comma Separated Values (CSV)

XML (eXtensible Markup Language)

JavaScript Object Notation (JSON)

```
id,name,amount,Remark
1,"Johnson, Smith, and Jones Co.",345.33,Pays on time
2,"Sam "Mad Dog" Smith",993.44,
3,"Barney & Company",0,"Great to work with and always
4,Johnson's Automotive,2344,
```


Data Formats

Plain Text

Comma Separated Values (CSV)

XML (eXtensible Markup Language)

JavaScript Object Notation (JSON)

```
<?xml version="1.0"?>
<ROWSET>
  <ROW>
    <id>1</id>
    <name>Johnson, Smith, and Jones Co.</name>
    <amount>345.33</amount>
    <Remark>Pays on time</Remark>
  </ROW>
  <ROW>
    <id>2</id>
    <name>Sam &quot;Mad Dog&quot; Smith</name>
    <amount>993.44</amount>
    <Remark></Remark>
  </ROW>
  <ROW>
    <id>3</id>
    <name>Barney &amp; Company</name>
    <amount>0</amount>
    <Remark>Great to work with and always pays w
  </ROW>
  <ROW>
    <id>4</id>
    <name>Johnson&apos;s Automotive</name>
    <amount>2344</amount>
    <Remark></Remark>
  </ROW>
</ROWSET>
```

Data Formats

Plain Text

Comma Separated Values (CSV)

XML (eXtensible Markup Language)

JavaScript Object Notation (JSON)

```
[
  {
    "id":1,
    "name":"Johnson, Smith, and Jones Co.",
    "amount":345.33,
    "Remark":"Pays on time"
  },
  {
    "id":2,
    "name":"Sam \"Mad Dog\" Smith",
    "amount":993.44,
    "Remark":""
  },
  {
    "id":3,
    "name":"Barney & Company",
    "amount":0,
    "Remark":"Great to work with and always pays withi
  },
  {
    "id":4,
    "name":"Johnson's Automotive",
    "amount":2344,
    "Remark":""
  }
]
```

Manipulation Tools

Don't underestimate Excel (comes with PowerQuery)

Commercial software (Tableau, [Tableau Public](#), Tableau Prep)

Programming/scripting languages: python (w/[beautiful soup](#)), R, Java, processing

Google tools ([Charts](#), [Fusion tables](#), [Ngram viewer](#), [API](#))

[OpenRefine](#)

[onlineocr.net](#)

[cloudconvert.org](#)

Break
until x:xx pm

Data Cleaning

Problematic Fields

Names!

Address information

Phone numbers, particularly if you need the area code to be separate

Date, if you only want year or month,

Time zones

Zip codes, can span other borders

Name		Middle Initials	Martina C. Daniels
Keith Pallard		Professional Designations	Lloyd Carson DVM
Fumi Takano		2-part Last Names	Lora de Carlo
Rhonda Johnson		Prefixes	Rev Herman Phillips
Warren Andersen		Suffixes	Jimmy Walford III
Juan Tyler		Hyphenated Last Names	Tori Baker-Andersen
Cicely Pope		Last Name First	Kincaid Jr, Paul
First Name		2-part First Names	Ray Anne Lipscomb
Last Name		Prefixes/Suffixes	Rev Rhonda-Lee St. Andrews-Fernandez, DD, MSW
	Keith Pallard	Other fields incorrectly included	Murray Wilkins 993 E Plymouth Blvd
Fumi Takano		No/Missing First Name	O'Connor
Rhonda	Johnson	No/Missing Last Name	Tanya
Warren Andersen		No name at all	
Juan	Tyler	I have no earthly idea!	JJ
Cicely	Pope	Not a person's name at all	North City Garden Supply

Cleaning Visualizing

Prep  **tableau**[®] Desktop
S O F T W A R E

 **TRIFACTA**

 **Power BI**

 **OpenRefine**



Excel Power Query

...and more!

Off-the-shelf vs. Roll your own



- + quick visual exploration
- + handling complex data
- limited flexibility

Final project should use more complex dataset



...and others

- + full design flexibility
- + nonstandard visualizations/interactions
- coding knowledge required

Final project can use simpler dataset

Some labs with Tableau

In canvas you will see THREE labs

**Do not panic. They are
short, but important.**

Next week: Additional Tableau lab

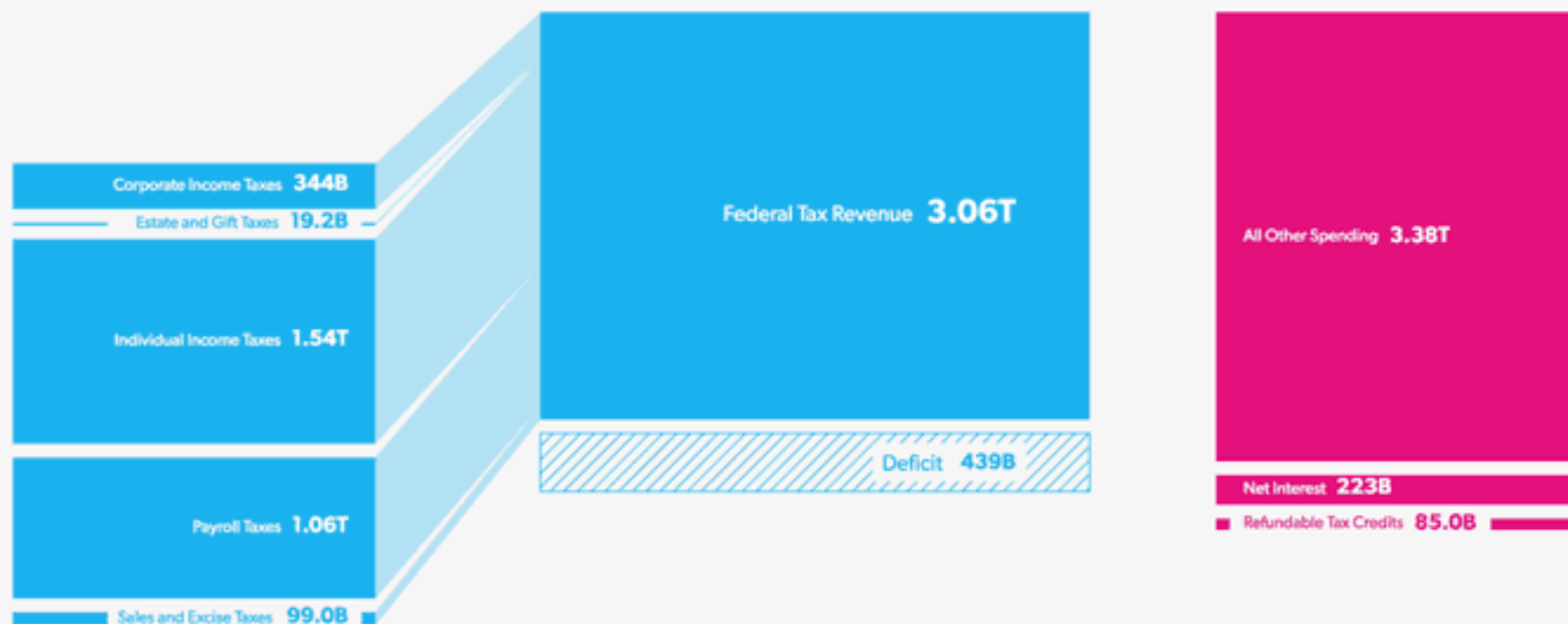
In the meantime:

<https://www.tableau.com/learn/training>

Why learn a tool?

So we can create helpful interactions for our users!

WHAT DO THE PROPOSED CHANGES MEAN FOR THE COUNTRY?



2015

Reports · Facts in Focus

Immigration

PURPOSE > WORK > GREEN CARD > EMPLOYMENT-BASED PREFERENCES

PROFESSIONALS WITH ADVANCED DEGREES OR ALIENS OF EXCEPTIONAL ABILITY

44K

PRIORITY WORKERS

42K

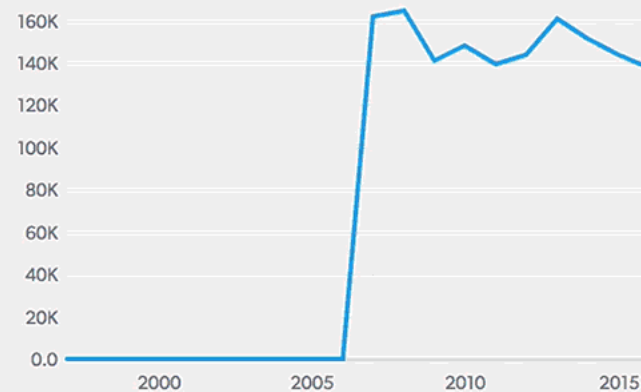
SKILLED WORKERS, PROFESSIONALS, AND UNSKILLED WORKERS

37K

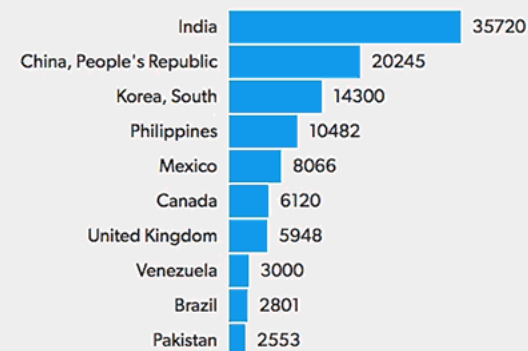
CERTAIN SPECIAL IMMIGRANTS

EMPLOYMENT CREATION (INVESTORS)

Number of immigrants throughout history



Top countries of origin





Quick Break
until x:xx pm

Assignment

Split your project group into 2 pairs*

Find a sufficiently complex data set relevant to your final project topic

Clean it

Transform it

Pose questions

Analyze the data

Document your process with images and comments

Project Groups

1

Liz Overbeek
Raquel Eisele
Julie Wang
Kavya Nagalakunta

2

Anoushka Halder
Tasnim Hasan
Ambika Ramesh Menon
Shreya Saxena

3

Jessica Hsu
Ralph Chang
Celia Tsao Trushaa
Ramanan

4

Amy La
MJ Schneider
Jacqueline Yan (Jackie)
Catherine Pan

5

Gily Yu
Sonali Arbune
Yiting Wang
Mai Kanapornchai

6

Ruiqi Chen
Amy Lee
Samantha Xiao
Honglam Van

7

Dina Chawla
Ko Motonaga
Jeremy Inducil
Julie Lee

Names are taken from student survey

Canvas Groups to be populated soon

What do you want to design and build?

What is the purpose of the visualization?

What results do you anticipate?

Goal is to get feedback on the scope

Talk to intended users about their needs!

What data you might use (more next week)

Early concepts and sketches

Project logistics

Tools

Project Proposal

Join up with your groups and discuss

Project proposal

Sourcing data

Your overall strategy

Group Time